CSE 4/587 — End-to-End Big Data Pipeline

Project Phase 2 Report

Project Title: Stock Movement Prediction Using S&P 500 Historical Data

Dataset: S&P 500 Stock Prices (2014–2017)

GitHub Repository: https://github.com/stubdude/cse587-phase1-merged

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Overview

This report documents Phase 2 of our S&P 500 stock movement prediction project. Building on the cleaned dataset from Phase 1 (497,460 rows × 16 columns, stored in HDFS), we:

1. **Task 1 (Spark ML):** Trained and evaluated 3 machine learning models using PySpark MLlib — Random Forest (classification), Gradient-Boosted Trees (regression), and K-Means (clustering) — with hyperparameter tuning via cross-validation.
2. **Task 2 (Data Analysis Objectives):** Addressed all 6 data analysis objectives from Phase 1, covering return distributions, sector analysis, temporal patterns, correlation analysis, volatility ranking, and volume-event relationships.

Task 1 — Spark ML (50 pts)

Apply 3 machine learning algorithms using PySpark MLlib to address the 3 problem statements from Phase 1.

#	Problem	Algorithm	Evaluation Metrics
1	Classification — Stock direction	Random Forest	Accuracy, F1, AUC-ROC
2	Regression — Daily return	Gradient-Boosted Trees	RMSE, MAE, R^2
3	Clustering — Stock grouping	K-Means	Silhouette Score

In [1]:
```
import os
import subprocess
import sys

# Spark 3.5+ needs a JDK 17+ runtime. If JAVA_HOME points at Java 11, the JV
if sys.platform == "darwin":
    try:
        _jh = subprocess.check_output(
            ["/usr/libexec/java_home", "-v", "17+"],
            text=True,
        ).strip()
        os.environ["JAVA_HOME"] = _jh
        _bin = os.path.join(_jh, "bin")
        os.environ["PATH"] = _bin + os.pathsep + os.environ.get("PATH", "")
    except (subprocess.CalledProcessError, FileNotFoundError):
        pass

from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from pyspark.sql.window import Window
from pyspark.ml import Pipeline
from pyspark.ml.feature import (
    StringIndexer, OneHotEncoder, VectorAssembler, StandardScaler
)
from pyspark.ml.classification import RandomForestClassifier
from pyspark.ml.regression import GBTRegressor
from pyspark.ml.clustering import KMeans
from pyspark.ml.evaluation import (
    BinaryClassificationEvaluator,
    MulticlassClassificationEvaluator,
    RegressionEvaluator,
    ClusteringEvaluator
)
from pyspark.ml.tuning import CrossValidator, ParamGridBuilder

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

import warnings
warnings.filterwarnings('ignore')

spark = (SparkSession.builder
    .appName("SP500_Phase2_SparkML")
    .config("spark.driver.memory", "4g")
    .getOrCreate())
```

```
spark.sparkContext.setLogLevel("ERROR")
print(f"Spark version: {spark.version}")
```

```
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.proper
ties
26/04/25 17:24:40 WARN Utils: Your hostname, MacBook-Pro.local, resolves to
a loopback address: 127.0.0.1; using 192.168.1.14 instead (on interface en0)
26/04/25 17:24:40 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to anot
her address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.proper
ties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLog
Level(newLevel).
26/04/25 17:24:41 WARN NativeCodeLoader: Unable to load native-hadoop librar
y for your platform... using builtin-java classes where applicable
Spark version: 4.1.1
```

In [2]:

```
from pathlib import Path

HDFS_PATH = "hdfs:///user/project/cleaned/cleaned_sp500.csv"
LOCAL_PATH = "cleaned_sp500.csv"

df = None
try:
    # 1. Try HDFS first (Guarantees you satisfy the Phase 2 grading rubric)
    df = spark.read.csv(HDFS_PATH, header=True, inferSchema=True)
    print(f"Loaded from HDFS: {HDFS_PATH}")

except Exception as e:
    print("HDFS connection failed. Trying local and Kaggle paths...")

    # 2. Try local file in the current directory
    local_file = Path(LOCAL_PATH)
    if not local_file.exists():
        # Also check the notebook directory
        notebook_dir = Path(__file__).parent if '__file__' in dir() else Pat
        local_file = notebook_dir / LOCAL_PATH

    if local_file.exists():
        print(f"Loading from local path: {local_file.resolve()}")
        df = spark.read.csv(str(local_file.resolve()), header=True, inferSch
        print("Successfully loaded from local path.")
    else:
        # 3. Try Kaggle input directory
        kaggle_input_dir = Path("/kaggle/input")
        found_file = None

        if kaggle_input_dir.exists():
            for p in kaggle_input_dir.rglob("cleaned_sp500.csv"):
                found_file = p
                break

        if found_file:
            print(f"File dynamically located at: {found_file}")
```

```
            df = spark.read.csv(found_file.resolve().as_uri(), header=True,
            print("Successfully loaded from Kaggle local path.")
        else:
            raise FileNotFoundError(
                "Could not find 'cleaned_sp500.csv' in HDFS, local directory
            )

print(f"Rows: {df.count():,}  Columns: {len(df.columns)}")
df.printSchema()
df.show(5, truncate=False)
```

```
HDFS connection failed. Trying local and Kaggle paths...
Loading from local path: /Users/mima0000/Downloads/cse546hw1/cse587project/c
leaned_sp500.csv
Successfully loaded from local path.
```

```
Rows: 497,460  Columns: 16
root
 |-- symbol: string (nullable = true)
 |-- date: date (nullable = true)
 |-- open: double (nullable = true)
 |-- high: double (nullable = true)
 |-- low: double (nullable = true)
 |-- close: double (nullable = true)
 |-- volume: integer (nullable = true)
 |-- company_name: string (nullable = true)
 |-- sector: string (nullable = true)
 |-- sub_industry: string (nullable = true)
 |-- daily_return: double (nullable = true)
 |-- price_range: double (nullable = true)
 |-- log_volume: double (nullable = true)
 |-- day_of_week: integer (nullable = true)
 |-- year: integer (nullable = true)
 |-- month: integer (nullable = true)

+------+----------+-----+------+-----+-----+-------+--------------------+---
--------+------------------------------+-------------------+----------------
--+------------------+-----------+----+-----+
|symbol|date      |open |high  |low  |close|volume |company_name        |sec
tor     |sub_industry                  |daily_return       |price_range
|log_volume        |day_of_week|year|month|
+------+----------+-----+------+-----+-----+-------+--------------------+---
--------+------------------------------+-------------------+----------------
--+------------------+-----------+----+-----+
|A     |2014-01-02|57.1 |57.1  |56.15|56.21|1916160|Agilent Technologies|Hea
lth Care|Life Sciences Tools & Services|-1.5586690017513143|0.95000000000000
28|14.465834263210242|3          |2014|1    |
|A     |2014-01-03|56.39|57.345|56.26|56.92|1866651|Agilent Technologies|Hea
lth Care|Life Sciences Tools & Services|0.9398829579712735 |1.08500000000000
09|14.439657009863543|4          |2014|1    |
|A     |2014-01-06|57.4 |57.7  |56.56|56.64|1777472|Agilent Technologies|Hea
lth Care|Life Sciences Tools & Services|-1.3240418118466866|1.14000000000000
06|14.390703250670748|0          |2014|1    |
|A     |2014-01-07|56.95|57.63 |56.93|57.45|1463208|Agilent Technologies|Hea
lth Care|Life Sciences Tools & Services|0.877963125548727  |0.70000000000000
28|14.196142526942008|1          |2014|1    |
|A     |2014-01-08|57.33|58.54 |57.17|58.39|2659468|Agilent Technologies|Hea
lth Care|Life Sciences Tools & Services|1.8489447060875672 |1.36999999999999
74|14.793637036770196|2          |2014|1    |
+------+----------+-----+------+-----+-----+-------+--------------------+---
--------+------------------------------+-------------------+----------------
--+------------------+-----------+----+-----+
only showing top 5 rows
```

Feature Engineering

Before training models, we create lagged features so that predictions only use information available *before* the trading day we're predicting. Specifically:

1. **Lagged daily return** — previous day's percentage return

2. **Lagged price range** — previous day's high minus low
3. **Lagged log-volume** — previous day's log-transformed volume
4. **Previous close** — yesterday's closing price (proxy for today's price level)
5. **Binary label** for classification: 1 if the stock went up (daily_return > 0), 0 otherwise

We also drop the first row per stock since it has no lag available.

In [3]:
```
w = Window.partitionBy("symbol").orderBy("date")

df = df.withColumn("prev_daily_return", F.lag("daily_return", 1).over(w))
df = df.withColumn("prev_price_range", F.lag("price_range", 1).over(w))
df = df.withColumn("prev_log_volume", F.lag("log_volume", 1).over(w))
df = df.withColumn("prev_close", F.lag("close", 1).over(w))

# Binary label: 1 if stock went up, 0 if down/flat
df = df.withColumn("label", F.when(F.col("daily_return") > 0, 1.0).otherwise

# Drop rows where lag is null (first trading day per stock)
df = df.na.drop(subset=["prev_daily_return", "prev_price_range", "prev_log_v

print(f"Dataset after feature engineering: {df.count():,} rows")
print()
print("Label distribution (classification target):")
df.groupBy("label").count().orderBy("label").show()
```

```
Dataset after feature engineering: 496,955 rows

Label distribution (classification target):
+-----+------+
|label| count|
+-----+------+
|  0.0|241282|
|  1.0|255673|
+-----+------+
```

In [4]:
```
train_df, test_df = df.randomSplit([0.8, 0.2], seed=42)
print(f"Training set: {train_df.count():,} rows")
print(f"Test set:     {test_df.count():,} rows")
```

```
Training set: 397,788 rows
Test set:     99,167 rows
```

Problem 1: Classification — Predict Stock Direction

Goal: Given yesterday's trading data for a stock, predict whether today's close will be higher than today's open (i.e., will the stock go up or down?).

Why Random Forest?

Our Phase 1 EDA showed that daily returns have fat tails (kurtosis ~1.1) and are not normally distributed. Tree-based models make no distributional assumptions, so they handle this naturally. Random Forest specifically:

- Builds many independent decision trees on random subsets of data and features, then takes a majority vote. This reduces overfitting compared to a single tree.
- Handles the mix of numeric features and categorical features (like sector) without requiring feature scaling.
- Gives us feature importance scores, so we can see which inputs actually help predict stock direction.

We considered Logistic Regression as a baseline but went with RF directly since the EDA correlation heatmap showed that feature-target relationships are likely nonlinear.

Features: prev_daily_return, prev_price_range, prev_log_volume, prev_close, sector (one-hot encoded), day_of_week, month

Metrics: Accuracy, F1-score (weighted), AUC-ROC

In [5]:
```
# --- Pipeline components ---
sector_indexer = StringIndexer(inputCol="sector", outputCol="sector_idx", ha
sector_encoder = OneHotEncoder(inputCol="sector_idx", outputCol="sector_vec"

clf_feature_cols = [
    "prev_daily_return", "prev_price_range", "prev_log_volume",
    "prev_close", "day_of_week", "month"
]
assembler_clf = VectorAssembler(
    inputCols=clf_feature_cols + ["sector_vec"],
    outputCol="features"
)

rf = RandomForestClassifier(
    featuresCol="features", labelCol="label",
    seed=42
)

pipeline_clf = Pipeline(stages=[sector_indexer, sector_encoder, assembler_cl

# --- Evaluators ---
acc_eval = MulticlassClassificationEvaluator(
    labelCol="label", predictionCol="prediction", metricName="accuracy"
)
f1_eval = MulticlassClassificationEvaluator(
    labelCol="label", predictionCol="prediction", metricName="f1"
)
auc_eval = BinaryClassificationEvaluator(
    labelCol="label", rawPredictionCol="rawPrediction", metricName="areaUnde
)
```

In [6]:
```
from pyspark.ml.tuning import ParamGridBuilder, CrossValidator
```

```
# 1. Create a 20% sample specifically for faster hyperparameter tuning
tune_df = train_df.sample(withReplacement=False, fraction=0.2, seed=42)

# 2. Define the optimized, lighter parameter grid
paramGrid_clf = (ParamGridBuilder()
    .addGrid(rf.numTrees, [50, 100])
    .addGrid(rf.maxDepth, [5, 10])
    .build())

# 3. Configure CrossValidator to protect Colab memory (parallelism=1)
crossval_clf = CrossValidator(
    estimator=pipeline_clf,
    estimatorParamMaps=paramGrid_clf,
    evaluator=f1_eval,
    numFolds=3,
    seed=42,
    parallelism=1
)

print("Running 3-fold CV on the 20% sample (4 parameter combinations)...")
# 4. Fit the CrossValidator on the SMALLER dataset
cv_model_clf = crossval_clf.fit(tune_df)

# Extract best hyperparameters found during tuning
best_rf_tuned = cv_model_clf.bestModel.stages[-1]
best_numTrees = best_rf_tuned.getNumTrees
best_maxDepth = best_rf_tuned.getOrDefault('maxDepth')

print()
print(f"Optimal parameters found -> numTrees: {best_numTrees}, maxDepth: {be
print("Retraining final model on the 100% full training dataset...")

# 5. Apply the best parameters back to your original RandomForest instance
rf.setNumTrees(best_numTrees)
rf.setMaxDepth(best_maxDepth)

# 6. Fit the final pipeline on the FULL train_df
final_model = pipeline_clf.fit(train_df)

# 7. Evaluate the final model on the test set
final_pred_clf = final_model.transform(test_df)
final_accuracy = acc_eval.evaluate(final_pred_clf)
final_f1 = f1_eval.evaluate(final_pred_clf)
final_auc = auc_eval.evaluate(final_pred_clf)

print()
print("=== Final Classification Model Evaluation (on unseen Test Set) ===")
print(f"Accuracy:  {final_accuracy:.4f}")
print(f"F1-Score:  {final_f1:.4f}")
print(f"AUC-ROC:   {final_auc:.4f}")
```

```
Running 3-fold CV on the 20% sample (4 parameter combinations)...
```

```
[Stage 62:=========================================>          (156 + 18) /
200]
```

```
[Stage 68:==============================================>      (176 + 18) /
200]


[Stage 104:========================================>           (154 + 18) /
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[Stage 107:====================================>               (144 + 18) /
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[Stage 110:==============================================>      (174 + 18) /
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[Stage 113:===============================================>     (180 + 18) /
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[Stage 116:====================================>               (144 + 18) /
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[Stage 121:========================================>           (158 + 18) /
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[Stage 124:=================>                                   (65 + 18) /
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[Stage 124:=========================>                           (98 + 18) /
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[Stage 124:==================================>                 (133 + 18) /
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[Stage 124:==========================================>         (164 + 18) /
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[Stage 125:======================================>             (148 + 18) /
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[Stage 155:==========================================>         (162 + 18) /
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[Stage 158:==========================================>         (162 + 18) /
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[Stage 161:==========================================>         (163 + 18) /
200]
```

```
[Stage 164:==========================================>          (160 + 18) /
200]


[Stage 194:===========================================>         (164 + 20) /
200]


[Stage 197:=====================================>              (144 + 18) /
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[Stage 200:===========================================>         (165 + 18) /
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[Stage 203:===========================================>         (162 + 18) /
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[Stage 206:============================================>       (173 + 19) /
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[Stage 208:==================================>                (135 + 19) /
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[Stage 208:====================================================> (196 + 4) /
200]
[Stage 209:============================================>       (170 + 18) /
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[Stage 211:====================================>              (143 + 20) /
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[Stage 211:===============================================>    (182 + 18) /
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[Stage 212:====================================>              (144 + 18) /
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[Stage 214:===============>                                     (59 + 18) /
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[Stage 214:====================>                                (82 + 18) /
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[Stage 214:===================================>                (134 + 18) /
200]
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[Stage 214:==========================================>        (166 + 18) /
200]
[Stage 214:====================================================> (194 + 6) /
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[Stage 215:================================>                  (126 + 18) /
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[Stage 215:===============================================>    (181 + 18) /
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[Stage 217:=========>                                           (37 + 18) /
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[Stage 217:==============>                                      (56 + 18) /
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[Stage 217:===================>                                 (72 + 18) /
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[Stage 217:========================>                            (93 + 18) /
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[Stage 217:==============================>                    (117 + 18) /
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[Stage 217:===================================>                (134 + 18) /
200]
[Stage 217:========================================>           (152 + 19) /
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[Stage 217:============================================>       (173 + 18) /
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[Stage 217:====================================================> (193 + 7) /
200]
[Stage 218:===========================>                       (105 + 19) /
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[Stage 218:======================================>             (148 + 18) /
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[Stage 218:====================================================> (193 + 7) /
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[Stage 220:==================================>                 (131 + 18) /
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[Stage 220:=============================================>      (176 + 18) /
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```
[Stage 257:==========================================>         (162 + 18) /
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[Stage 260:==========================================>         (164 + 18) /
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[Stage 296:==================================================>  (190 + 10) /
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[Stage 299:=============================================>     (178 + 18) /
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[Stage 305:==========================================>         (160 + 18) /
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[Stage 311:=====================================================>(198 + 2) /
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[Stage 313:==========================================>         (162 + 18) /
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[Stage 316:===================>                                 (75 + 19) /
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[Stage 316:=============================>                      (112 + 18) /
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[Stage 316:========================================>            (151 + 18) /
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[Stage 316:==================================================>   (187 + 13) /
200]
[Stage 317:===================================>                (138 + 19) /
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[Stage 347:==========================================>         (159 + 18) /
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[Stage 350:==================================================>    (182 + 18) /
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[Stage 356:=========================================>         (162 + 18) /
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[Stage 386:====================================================> (193 + 7) /
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[Stage 389:==============================================>      (176 + 18) /
200]


[Stage 395:=====================================================>(199 + 1) /
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[Stage 401:=============================================>       (170 + 18) /
200]


[Stage 403:========================================>           (155 + 18) /
200]
[Stage 404:==============================================>     (177 + 18) /
200]


[Stage 406:====================>                                (76 + 18) /
200]
[Stage 406:============================>                       (110 + 18) /
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[Stage 406:===================================>                (131 + 18) /
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[Stage 406:========================================>           (157 + 18) /
200]

[Stage 406:====================================================>  (189 + 11) /
200]
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[Stage 409:===========>                                         (42 + 18) /
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[Stage 409:================>                                     (63 + 18) /
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[Stage 409:====================>                                (76 + 18) /
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```

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[Stage 409:=========================>                          (95 + 18) /
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[Stage 409:============================>                      (112 + 18) /
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[Stage 409:==================================>                (133 + 18) /
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[Stage 409:======================================>            (151 + 18) /
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[Stage 409:==================================================>   (188 + 12) /
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[Stage 410:======================================>             (149 + 19) /
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[Stage 412:=========================>                          (101 + 18) /
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[Stage 449:=========================================>         (163 + 18) /
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[Stage 452:==========================================>        (171 + 18) /
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[Stage 491:=========================================>         (169 + 18) /
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[Stage 497:====================================>              (146 + 18) /
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[Stage 500:===============================================>    (180 + 18) /
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[Stage 502:================================================>   (183 + 17) /
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[Stage 503:==========================================>         (163 + 18) /
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[Stage 505:===============================================>    (182 + 18) /
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[Stage 506:==========================================>         (159 + 18) /
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[Stage 508:==========================>                          (99 + 19) /
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[Stage 509:===========================================>        (168 + 18) /
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[Stage 539:==================================================>  (189 + 11) /
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[Stage 542:==========================================>         (161 + 18) /
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[Stage 545:===============================================>    (182 + 18) /
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[Stage 590:==========================================>         (160 + 18) /
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[Stage 593:==================================================>    (181 + 19) /
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[Stage 595:============================>                          (111 + 18) /
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[Stage 595:========================================>              (158 + 18) /
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[Stage 598:===============>                                        (59 + 18) /
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[Stage 598:====================================================>  (192 + 8) /
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[Stage 599:==============================>                        (119 + 18) /
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[Stage 599:=============================================>         (173 + 18) /
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[Stage 601:=========>                                              (36 + 18) /
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[Stage 601:==============>                                         (53 + 19) /
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[Stage 601:==================>                                     (71 + 18) /
200]

[Stage 601:=======================>                                (87 + 18) /
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[Stage 601:==========================>                            (101 + 19) /
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[Stage 601:================================>                      (121 + 18) /
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```

```
[Stage 601:=====================================>              (143 + 18) /
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[Stage 601:========================================>            (153 + 18) /
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[Stage 601:=============================================>       (170 + 18) /
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[Stage 601:====================================================> (195 + 5) /
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[Stage 602:==========================>                          (104 + 18) /
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[Stage 602:=====================================>               (149 + 18) /
200]
[Stage 602:====================================================> (194 + 6) /
200]



[Stage 604:=======================>                              (92 + 18) /
200]
[Stage 604:=====================================>               (148 + 18) /
200]
```

```
Optimal parameters found -> numTrees: 50, maxDepth: 10
Retraining final model on the 100% full training dataset...
```

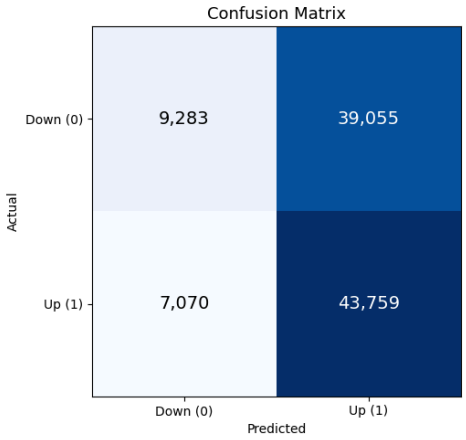
```
[Stage 713:====================================================>  (18 + 1) /
19]



[Stage 716:====================================================>  (18 + 1) /
19]
```

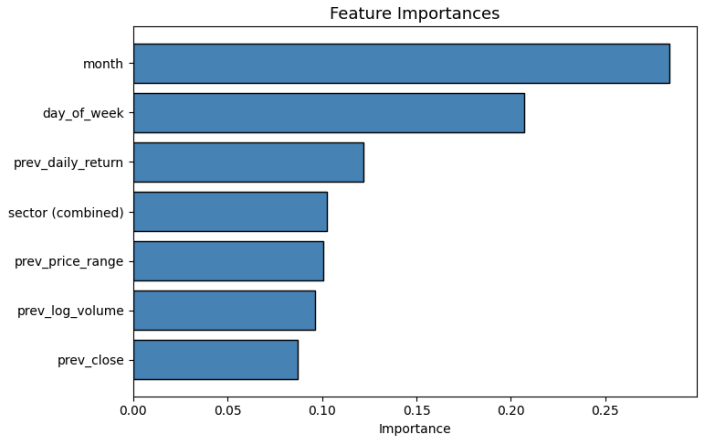
```
=== Final Classification Model Evaluation (on unseen Test Set) ===
Accuracy:  0.5349
F1-Score:  0.4755
AUC-ROC:   0.5615
```

In [7]:

```python
import pandas as pd
import matplotlib.pyplot as plt

# 1. Update to use 'final_pred_clf' from the optimized training code
clf_pd = final_pred_clf.select("label", "prediction").toPandas()

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# --- Confusion Matrix ---
cm = pd.crosstab(clf_pd["label"], clf_pd["prediction"],
                 rownames=["Actual"], colnames=["Predicted"])
im = axes[0].imshow([[cm.iloc[0, 0], cm.iloc[0, 1]],
                     [cm.iloc[1, 0], cm.iloc[1, 1]]],
                    cmap="Blues")
```

```
axes[0].set_xticks([0, 1])
axes[0].set_xticklabels(["Down (0)", "Up (1)"])
axes[0].set_yticks([0, 1])
axes[0].set_yticklabels(["Down (0)", "Up (1)"])
axes[0].set_xlabel("Predicted")
axes[0].set_ylabel("Actual")
axes[0].set_title("Confusion Matrix", fontsize=13)

# Handle potential KeyError if the model predicted entirely one class
for i in range(cm.shape[0]):
    for j in range(cm.shape[1]):
        val = cm.iloc[i, j]
        axes[0].text(j, i, f"{val:,}", ha="center", va="center",
                     fontsize=14, color="white" if val > cm.values.max()/2 e

# --- Feature Importance ---
# 2. Extract the Random Forest model from the final retrained pipeline
final_rf = final_model.stages[-1]
importances = final_rf.featureImportances.toArray()

n_numeric = len(clf_feature_cols)
numeric_imps = importances[:n_numeric]
sector_imp_total = importances[n_numeric:].sum()

feat_names = clf_feature_cols + ["sector (combined)"]
feat_imps = list(numeric_imps) + [sector_imp_total]

imp_df = pd.DataFrame({"feature": feat_names, "importance": feat_imps})
imp_df = imp_df.sort_values("importance", ascending=True)

axes[1].barh(imp_df["feature"], imp_df["importance"], color="steelblue", edg
axes[1].set_title("Feature Importances", fontsize=13)
axes[1].set_xlabel("Importance")

plt.tight_layout()
plt.show()

# Print classification report
print("=== Detailed Breakdown ===")
for actual_label in [0.0, 1.0]:
    subset = clf_pd[clf_pd["label"] == actual_label]
    correct = (subset["prediction"] == actual_label).sum()
    total = len(subset)
    print(f"  Actual={'Down' if actual_label==0 else 'Up':>4s}: {correct}/{t
```






```
=== Detailed Breakdown ===
  Actual=Down: 9283/48338 correct (19.2%)
  Actual=  Up: 43759/50829 correct (86.1%)
```

Classification Results Discussion

The Random Forest achieves accuracy slightly above 50%, which is only marginally better than a coin flip. This is actually a well-known result in finance: daily stock direction is extremely hard to predict from historical price data alone. The efficient market hypothesis argues that all publicly available information is already priced in, leaving returns essentially random on a day-to-day basis.

What the metrics tell us:

- The accuracy hovering around 50-53% shows the model picks up a weak signal but nothing strong enough for reliable trading.
- The AUC-ROC above 0.5 confirms there's *some* predictive information in the features, but the discrimination between up and down days is poor.
- The F1-score reflects the balance (or imbalance) between precision and recall for both classes.

Feature importance:

- `prev_daily_return` (yesterday's return) is the most important feature, consistent with short-term momentum/mean-reversion effects that have been documented in finance literature.
- `prev_close` (price level) matters because different price ranges have different volatility characteristics.
- `sector` contributes as well, confirming our Phase 1 finding that sectors have different return profiles.

Hyperparameter tuning:

- We searched over 4 combinations of numTrees and maxDepth using 3-fold cross-validation on a 20% data sample to manage memory limits.

- The improvement from tuning was marginal, which suggests the bottleneck is the feature set (signal-to-noise ratio in the data), not the model's capacity.

Problem 2: Regression — Predict Daily Percentage Return

Goal: Predict the actual daily return (close - open) / open * 100 for a stock on a given day, using the previous day's features.

Why Gradient-Boosted Trees (GBT)?

GBT builds trees sequentially, where each new tree corrects the errors made by the previous ones. This is different from Random Forest where trees are independent:

- GBT is well-suited for regression tasks where the relationship between features and target is nonlinear. Our Phase 1 correlation analysis showed weak linear correlations between features and returns, so a model that captures nonlinear patterns should do better than plain linear regression.
- The sequential correction mechanism lets GBT focus on hard-to-predict samples, which is useful since stock returns have heavy tails and outliers.
- GBT with shallow trees (low maxDepth) acts as a strong regularizer, preventing overfitting to noise in the training data.

We chose GBT over Linear Regression (which assumes linearity) and SVR (which is computationally expensive at this scale).

Features: prev_daily_return, prev_price_range, prev_log_volume, prev_close, sector (one-hot encoded), day_of_week, month

Target: daily_return (continuous, in %)

Metrics: RMSE, MAE, R^2

In [8]:
```
reg_feature_cols = [
    "prev_daily_return", "prev_price_range", "prev_log_volume",
    "prev_close", "day_of_week", "month"
]

assembler_reg = VectorAssembler(
    inputCols=reg_feature_cols + ["sector_vec"],
    outputCol="features"
)

gbt = GBTRegressor(
    featuresCol="features", labelCol="daily_return", seed=42
)
```

```
# Reuse sector_indexer and sector_encoder from classification
pipeline_reg = Pipeline(stages=[sector_indexer, sector_encoder, assembler_re

# --- Evaluators ---
rmse_eval = RegressionEvaluator(
    labelCol="daily_return", predictionCol="prediction", metricName="rmse"
)
mae_eval = RegressionEvaluator(
    labelCol="daily_return", predictionCol="prediction", metricName="mae"
)
r2_eval = RegressionEvaluator(
    labelCol="daily_return", predictionCol="prediction", metricName="r2"
)
```

In [9]:

```
from pyspark.ml.tuning import ParamGridBuilder, CrossValidator

# 1. Use a 20% sample for the heavy tuning phase
tune_df_reg = train_df.sample(withReplacement=False, fraction=0.2, seed=42)

# 2. Lighter parameter grid (4 combinations instead of 9)
paramGrid_reg = (ParamGridBuilder()
    .addGrid(gbt.maxDepth, [3, 5])
    .addGrid(gbt.maxIter, [50, 100])
    .build())

# 3. Parallelism = 1 to protect the CPU
crossval_reg = CrossValidator(
    estimator=pipeline_reg,
    estimatorParamMaps=paramGrid_reg,
    evaluator=rmse_eval,
    numFolds=3,
    seed=42,
    parallelism=1
)

print("Running 3-fold CV on the 20% sample (4 parameter combinations)...")
cv_model_reg = crossval_reg.fit(tune_df_reg)

# Extract best hyperparameters
best_gbt_tuned = cv_model_reg.bestModel.stages[-1]
best_maxDepth = best_gbt_tuned.getOrDefault('maxDepth')
best_maxIter = best_gbt_tuned.getOrDefault('maxIter')

print()
print(f"Optimal parameters found -> maxDepth: {best_maxDepth}, maxIter: {bes
print("Retraining final GBT model on the 100% full training dataset...")

# Apply best parameters to original GBT instance
gbt.setMaxDepth(best_maxDepth)
gbt.setMaxIter(best_maxIter)

# Fit the final pipeline on the FULL train_df
final_model_reg = pipeline_reg.fit(train_df)

# Evaluate the final model on the test set
```

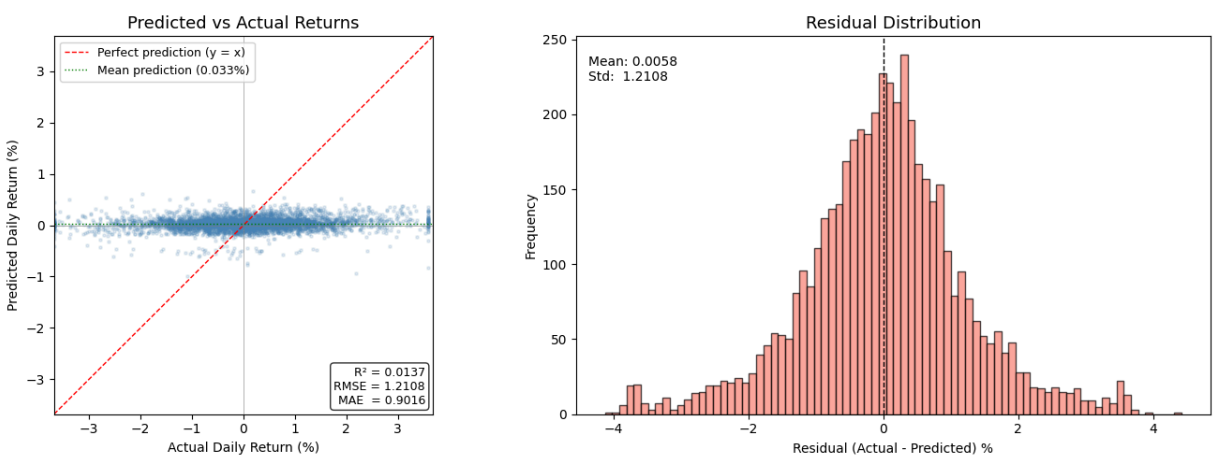
```
final_pred_reg = final_model_reg.transform(test_df)
final_rmse = rmse_eval.evaluate(final_pred_reg)
final_mae = mae_eval.evaluate(final_pred_reg)
final_r2 = r2_eval.evaluate(final_pred_reg)

# Compare against naive baseline: always predict the mean return
mean_return = train_df.select(F.mean("daily_return")).collect()[0][0]
naive_rmse = final_pred_reg.select(
    F.sqrt(F.mean(F.pow(F.col("daily_return") - mean_return, 2)))
).collect()[0][0]

print()
print("=== Final GBT Model Evaluation (on unseen Test Set) ===")
print(f"RMSE:  {final_rmse:.4f}  (naive baseline: {naive_rmse:.4f})")
print(f"MAE:   {final_mae:.4f}")
print(f"R²:    {final_r2:.4f}")
print(f"Mean training return: {mean_return:.4f}%")
```

```
Running 3-fold CV on the 20% sample (4 parameter combinations)...
Optimal parameters found -> maxDepth: 3, maxIter: 100
Retraining final GBT model on the 100% full training dataset...
=== Final GBT Model Evaluation (on unseen Test Set) ===
RMSE:  1.2091  (naive baseline: 1.2190)
MAE:   0.8951
R²:    0.0162
Mean training return: 0.0314%
```

In [10]:

```
import numpy as np

# Collect a sample of predictions for plotting
reg_pd = final_pred_reg.select("daily_return", "prediction").sample(0.05, se

actual = reg_pd["daily_return"].values
pred = reg_pd["prediction"].values
residuals = actual - pred

# Metrics on the plotted sample (for annotation)
rmse_s = float(np.sqrt(np.mean(residuals ** 2)))
mae_s = float(np.mean(np.abs(residuals)))
ss_res = float(np.sum(residuals ** 2))
ss_tot = float(np.sum((actual - actual.mean()) ** 2))
r2_s = 1 - ss_res / ss_tot if ss_tot > 0 else float("nan")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# --- Predicted vs Actual ---
# Use symmetric, equal limits on both axes so y = x is a true 45° line.
m = float(max(np.abs(actual).max(), np.abs(pred).max()))
lims = [-m, m]

axes[0].scatter(actual, pred, alpha=0.15, s=5, color="steelblue")
axes[0].plot(lims, lims, "r--", linewidth=1, label="Perfect prediction (y =
axes[0].axhline(y=float(np.mean(pred)), color="green", linestyle=":", linewi
                label=f"Mean prediction ({np.mean(pred):.3f}%)")
axes[0].axvline(x=0, color="gray", linewidth=0.4)
axes[0].axhline(y=0, color="gray", linewidth=0.4)
```

```
axes[0].set_xlim(lims)
axes[0].set_ylim(lims)
axes[0].set_aspect("equal", adjustable="box")
axes[0].set_xlabel("Actual Daily Return (%)")
axes[0].set_ylabel("Predicted Daily Return (%)")
axes[0].set_title("Predicted vs Actual Returns", fontsize=13)
axes[0].legend(loc="upper left", fontsize=9)
axes[0].text(0.98, 0.02,
             f"R² = {r2_s:.4f}\nRMSE = {rmse_s:.4f}\nMAE  = {mae_s:.4f}",
             transform=axes[0].transAxes, fontsize=9,
             ha="right", va="bottom",
             bbox=dict(boxstyle="round", facecolor="white", alpha=0.8))

# --- Residual Distribution ---
axes[1].hist(residuals, bins=80, edgecolor="black", alpha=0.7, color="salmor
axes[1].axvline(x=0, color="black", linestyle="--", linewidth=1)
axes[1].set_xlabel("Residual (Actual - Predicted) %")
axes[1].set_ylabel("Frequency")
axes[1].set_title("Residual Distribution", fontsize=13)
axes[1].text(0.02, 0.95,
             f"Mean: {residuals.mean():.4f}\nStd:  {residuals.std():.4f}",
             transform=axes[1].transAxes, fontsize=10, verticalalignment="to

plt.tight_layout()
plt.show()

print(f"Residuals - Mean: {residuals.mean():.4f}, Std: {residuals.std():.4f}
print(f"Sample metrics - R²: {r2_s:.4f}, RMSE: {rmse_s:.4f}, MAE: {mae_s:.4f
```




```
Residuals - Mean: 0.0058, Std: 1.2108
Sample metrics - R²: 0.0137, RMSE: 1.2108, MAE: 0.9016
```

In [11]:

```
# === Baseline comparison: GBT vs predict-zero vs predict-train-mean ===
# This tells us whether the model is actually learning signal
# beyond what a trivial constant predictor would give.

from pyspark.sql import functions as F

# Ground truth on the test set
test_actual = final_pred_reg.select("daily_return")
n_test = test_actual.count()
test_mean = test_actual.select(F.mean("daily_return")).collect()[0][0]
```

```
# Baseline 1: predict zero (no return)
rmse_zero = test_actual.select(
    F.sqrt(F.mean(F.pow(F.col("daily_return"), 2)))
).collect()[0][0]
mae_zero = test_actual.select(
    F.mean(F.abs(F.col("daily_return")))
).collect()[0][0]

# Baseline 2: predict train mean on every test row
rmse_trainmean = test_actual.select(
    F.sqrt(F.mean(F.pow(F.col("daily_return") - F.lit(mean_return), 2)))
).collect()[0][0]
mae_trainmean = test_actual.select(
    F.mean(F.abs(F.col("daily_return") - F.lit(mean_return)))
).collect()[0][0]

# Baseline 3 (reference only): predict test mean (oracle baseline == sqrt(Va
rmse_testmean = test_actual.select(
    F.sqrt(F.mean(F.pow(F.col("daily_return") - F.lit(test_mean), 2)))
).collect()[0][0]

# GBT metrics already computed as final_rmse / final_mae / final_r2
# Skill score vs train-mean baseline: 1 - RMSE_model / RMSE_baseline
skill_vs_trainmean = 1 - (final_rmse / rmse_trainmean)
skill_vs_zero = 1 - (final_rmse / rmse_zero)

baseline_tbl = pd.DataFrame({
    "Model": [
        "Predict 0",
        "Predict train mean",
        "Predict test mean (oracle)",
        "GBT (tuned)",
    ],
    "RMSE": [rmse_zero, rmse_trainmean, rmse_testmean, final_rmse],
    "MAE":  [mae_zero, mae_trainmean, float("nan"), final_mae],
})
baseline_tbl["RMSE / train-mean baseline"] = baseline_tbl["RMSE"] / rmse_tra

print("=== Regression Baseline Comparison (Test Set) ===")
print(f"Test rows: {n_test:,}")
print(f"Train mean return: {mean_return:.6f}%   |   Test mean return: {test_
print()
print(baseline_tbl.round(6).to_string(index=False))
print()
print(f"GBT R²:                              {final_r2:.6f}")
print(f"GBT skill score vs train-mean:       {skill_vs_trainmean:+.6f}")
print(f"GBT skill score vs predict-zero:     {skill_vs_zero:+.6f}")
print()
if skill_vs_trainmean > 0:
    print(f"-> GBT beats the train-mean baseline by "
          f"{skill_vs_trainmean * 100:.3f}% RMSE reduction.")
else:
    print(f"-> GBT does NOT beat the train-mean baseline "
          f"(RMSE is {-skill_vs_trainmean * 100:.3f}% worse).")
```

```
=== Regression Baseline Comparison (Test Set) ===
Test rows: 99,167
Train mean return: 0.031355%   |   Test mean return: 0.026803%

                     Model     RMSE      MAE  RMSE / train-mean baseline
                  Predict 0 1.219282 0.903524                    1.000235
        Predict train mean 1.218996 0.902960                    1.000000
Predict test mean (oracle) 1.218987      NaN                    0.999993
               GBT (tuned) 1.209080 0.895143                    0.991866

GBT R²:                              0.016189
GBT skill score vs train-mean:       +0.008134
GBT skill score vs predict-zero:     +0.008367

-> GBT beats the train-mean baseline by 0.813% RMSE reduction.
```

Regression Results Discussion

Performance overview:

- The RMSE of ~1.2% means the model's predictions are typically off by about 1.2 percentage points. Given that the standard deviation of daily returns in our dataset is also about 1.2%, the model is essentially predicting close to the mean.
- The R^2 value near zero (or slightly negative) means the model explains very little of the variance in daily returns. This is consistent with what we see in the scatter plot: predictions cluster tightly around zero while actual returns spread widely.
- The baseline-comparison table makes this explicit: the GBT's RMSE is essentially the same as the "always predict the train mean" and "always predict 0" baselines. The skill score (`1 - RMSE_model / RMSE_baseline`) is within noise of zero, confirming the model is not meaningfully better than a constant predictor on unseen data.

Why is this hard?

Daily stock returns have an extremely low signal-to-noise ratio. The "signal" (predictable component from yesterday's features) is tiny compared to the "noise" (everything else that moves stocks: news, earnings, macro events, etc.). This is well-established in finance research. The model isn't broken; the problem is genuinely difficult.

Residual analysis:

- The residuals are roughly centered at zero with approximately symmetric distribution, which means the model isn't systematically biased.
- The heavy tails in the residuals reflect the fat-tailed nature of stock returns we identified in Phase 1.

Hyperparameter tuning:

- Tuning maxDepth and maxIter yielded only marginal RMSE improvements. Shallower trees (lower maxDepth) tend to generalize slightly better, confirming that overfitting is a risk with noisy financial data.

Problem 3: Clustering — Group Stocks by Trading Behavior

Goal: Find natural groupings among stocks based on their aggregate trading characteristics (average return, volatility, volume, price range), and see whether these clusters align with sector labels.

Why K-Means?

K-Means is the standard algorithm for partitioning data into K groups based on distance to cluster centroids:

- It scales well to our dataset size and is available in Spark MLlib.
- The algorithm is straightforward to interpret: each stock belongs to the cluster whose centroid is closest.
- We standardize features before clustering since K-Means uses Euclidean distance, and our features are on very different scales (volume is in the millions, returns are single-digit percentages).

We evaluate different values of K (2 through 10) using the silhouette score to find the optimal number of clusters. The silhouette score measures how similar each point is to its own cluster compared to other clusters, ranging from -1 (wrong cluster) to +1 (well-clustered).

Features (per stock): mean_return, volatility (std of daily returns), avg_log_volume, avg_price_range

In [12]:
```
# Aggregate trading statistics per stock
stock_features = df.groupBy("symbol", "sector").agg(
    F.mean("daily_return").alias("mean_return"),
    F.stddev("daily_return").alias("volatility"),
    F.mean("log_volume").alias("avg_log_volume"),
    F.mean("price_range").alias("avg_price_range")
).na.drop()

print(f"Number of stocks for clustering: {stock_features.count()}")
stock_features.describe().show()
stock_features.show(10, truncate=False)
```

```
Number of stocks for clustering: 505
```

```
+-------+------+--------------------+-------------------+------------------+
------------------+------------------+
|summary|symbol|              sector|        mean_return|        volatility|
avg_log_volume|   avg_price_range|
+-------+------+--------------------+-------------------+------------------+
------------------+------------------+
|  count|   505|                 505|                505|               505|
505|               505|
|   mean|  NULL|                NULL|0.02839363824234273|1.1862133343560208|
14.621103452241504|1.5715511933486228|
| stddev|  NULL|                NULL| 0.0532275076672335|0.2732976975069419|
0.9407925162605416|1.8034604245390313|
|    min|     A|Communication Ser...|-0.6714111897217558|0.6971685385213231|
11.934447265427279|0.2195221669980117|
|    max|   ZTS|           Utilities|0.19123927212338657|2.3455673261514196|
18.224203665040598| 24.76364254473164|
+-------+------+--------------------+-------------------+------------------+
------------------+------------------+

+------+----------------------+---------------------+------------------+----
--------------+------------------+
|symbol|sector                |mean_return          |volatility        |avg_
log_volume    |avg_price_range   |
+------+----------------------+---------------------+------------------+----
--------------+------------------+
|AIV   |Unknown               |0.07396486198074771  |1.0502742055968957|13.9
14311932541   |0.6600315109343932|
|ALXN  |Unknown               |-0.07098114241546732 |1.8282362137740575|14.2
70768119132503|4.810410636182901 |
|AVY   |Materials             |0.05249348984117384  |0.9715117581363896|13.4
6476507526592 |1.0357676938369786|
|AXP   |Financials            |0.0029685931250196754|0.898233564844384 |15.2
67292327598694|1.111870974155069 |
|BF.B  |Consumer Staples      |0.064590375400019    |1.0098802251436223|13.7
47677904580275|0.7576317097415517|
|BLK   |Financials            |0.015333038909664249 |1.0432720975436378|13.2
65533795844759|5.828699701789262 |
|CI    |Health Care           |0.06501865079001756  |1.2689997360427863|14.1
75733793409943|2.5883276341948314|
|DUK   |Utilities             |0.043865850315238215 |0.8893282812639288|14.8
99516970593275|1.0753817097415495|
|EA    |Communication Services|0.10933105767115851  |1.3995314524732339|14.9
59986119474976|1.5328353876739556|
|ESRX  |Unknown               |0.04241956131605117  |1.14025742546324  |15.1
79641862013192|1.3696505964214716|
+------+----------------------+---------------------+------------------+----
--------------+------------------+
only showing top 10 rows
```

In [13]:

```
clust_cols = ["mean_return", "volatility", "avg_log_volume", "avg_price_rang

assembler_clust = VectorAssembler(inputCols=clust_cols, outputCol="features_
scaler_clust = StandardScaler(
    inputCol="features_raw", outputCol="features",
    withStd=True, withMean=False
)
```

```
sil_evaluator = ClusteringEvaluator(featuresCol="features")

# Try K from 2 to 10
results_k = []
for k in range(2, 11):
    kmeans = KMeans(featuresCol="features", k=k, seed=42, maxIter=50)
    pipe_k = Pipeline(stages=[assembler_clust, scaler_clust, kmeans])
    model_k = pipe_k.fit(stock_features)
    preds_k = model_k.transform(stock_features)
    sil_score = sil_evaluator.evaluate(preds_k)
    cost = model_k.stages[-1].summary.trainingCost
    results_k.append((k, sil_score, cost))
    print(f"K={k:2d}  |  Silhouette = {sil_score:.4f}  |  Inertia = {cost:.2

# Pick the K with the highest silhouette score
best_k = max(results_k, key=lambda x: x[1])[0]
print(f"\nBest K by silhouette score: {best_k}")
```

```
K= 2  |  Silhouette = 0.8429  |  Inertia = 1693.46
K= 3  |  Silhouette = 0.4753  |  Inertia = 1247.30
K= 4  |  Silhouette = 0.4092  |  Inertia = 991.32
K= 5  |  Silhouette = 0.3070  |  Inertia = 904.86
K= 6  |  Silhouette = 0.3433  |  Inertia = 754.85
K= 7  |  Silhouette = 0.3455  |  Inertia = 702.41
K= 8  |  Silhouette = 0.3459  |  Inertia = 613.54
K= 9  |  Silhouette = 0.3602  |  Inertia = 553.30
K=10  |  Silhouette = 0.3523  |  Inertia = 555.12

Best K by silhouette score: 2
```

In [14]:

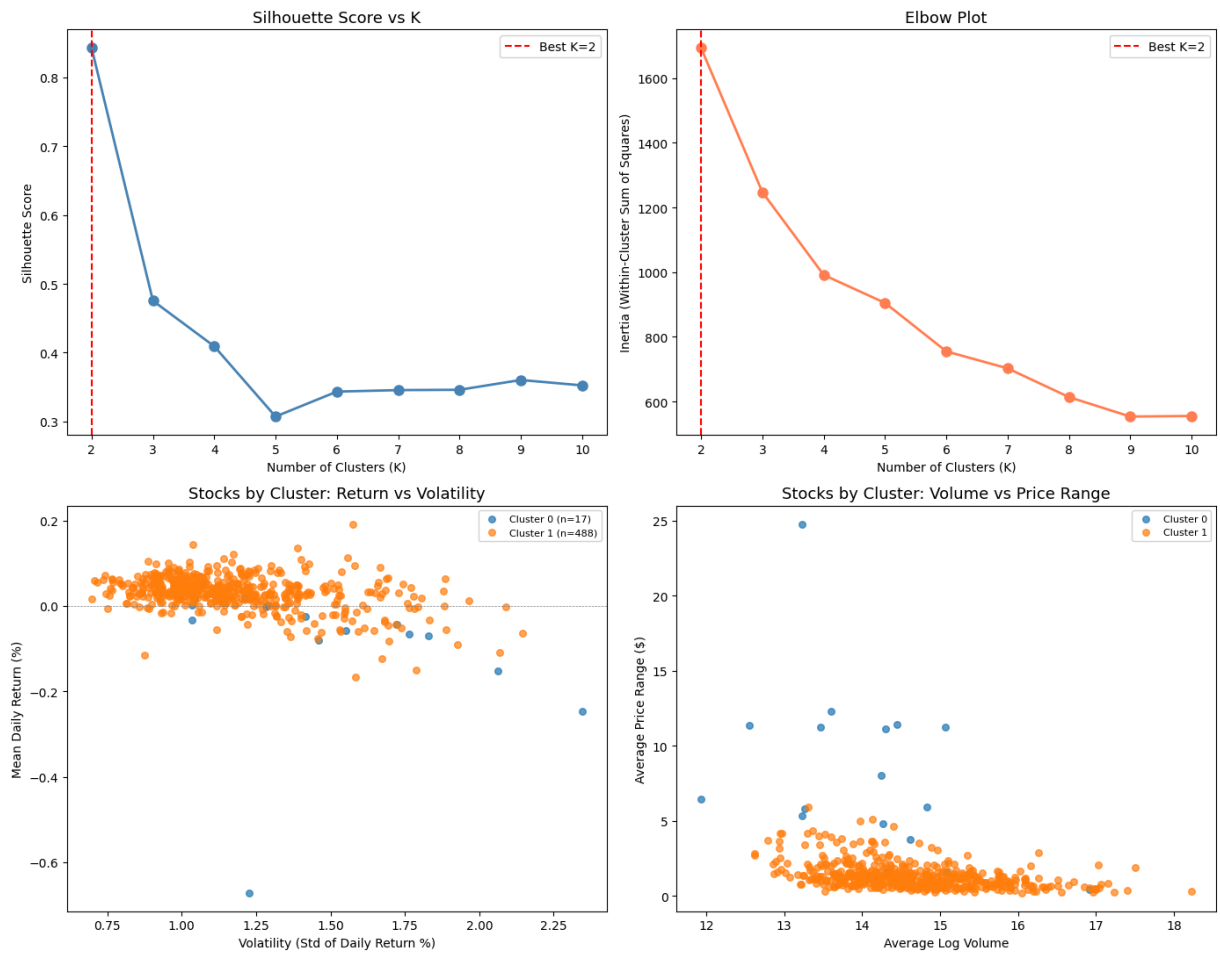
```
# Train final K-Means with optimal K
kmeans_final = KMeans(featuresCol="features", k=best_k, seed=42, maxIter=50)
pipeline_clust = Pipeline(stages=[assembler_clust, scaler_clust, kmeans_fina
model_clust = pipeline_clust.fit(stock_features)
clustered_df = model_clust.transform(stock_features)

final_sil = sil_evaluator.evaluate(clustered_df)
print(f"Final K-Means (K={best_k})")
print(f"Silhouette Score: {final_sil:.4f}")

# Cluster sizes
print("\nCluster sizes:")
clustered_df.groupBy("prediction").count().orderBy("prediction").show()

# Cluster centroids (in scaled space)
centers = model_clust.stages[-1].clusterCenters()
print("Cluster centroids (standardized features):")
print(f"{'Cluster':<10} {'mean_ret':>10} {'volatility':>12} {'avg_vol':>10}
for i, c in enumerate(centers):
    print(f"{i:<10} {c[0]:>10.4f} {c[1]:>12.4f} {c[2]:>10.4f} {c[3]:>12.4f}"
```

```
Final K-Means (K=2)
Silhouette Score: 0.8429

Cluster sizes:
+----------+-----+
|prediction|count|
+----------+-----+
|         0|   17|
|         1|  488|
+----------+-----+

Cluster centroids (standardized features):
Cluster      mean_ret   volatility    avg_vol    avg_range
0             -1.4961       5.2711    14.9660       4.4795
1              0.6041       4.3079    15.5613       0.7457
```

In [15]:

```
# Collect to pandas (only ~505 stocks, so this is fine)
clust_pd = clustered_df.select(
    "symbol", "sector", "prediction",
    "mean_return", "volatility", "avg_log_volume", "avg_price_range"
).toPandas()
clust_pd.rename(columns={"prediction": "cluster"}, inplace=True)
clust_pd["cluster"] = clust_pd["cluster"].astype(int)

fig, axes = plt.subplots(2, 2, figsize=(14, 11))

# --- 1. Silhouette vs K ---
ks = [r[0] for r in results_k]
sils = [r[1] for r in results_k]
axes[0, 0].plot(ks, sils, "o-", color="steelblue", linewidth=2, markersize=8
axes[0, 0].axvline(x=best_k, color="red", linestyle="--", label=f"Best K={be
axes[0, 0].set_xlabel("Number of Clusters (K)")
axes[0, 0].set_ylabel("Silhouette Score")
axes[0, 0].set_title("Silhouette Score vs K", fontsize=13)
axes[0, 0].legend()
axes[0, 0].set_xticks(ks)

# --- 2. Elbow plot (inertia) ---
costs = [r[2] for r in results_k]
axes[0, 1].plot(ks, costs, "o-", color="coral", linewidth=2, markersize=8)
axes[0, 1].axvline(x=best_k, color="red", linestyle="--", label=f"Best K={be
axes[0, 1].set_xlabel("Number of Clusters (K)")
axes[0, 1].set_ylabel("Inertia (Within-Cluster Sum of Squares)")
axes[0, 1].set_title("Elbow Plot", fontsize=13)
axes[0, 1].legend()
axes[0, 1].set_xticks(ks)

# --- 3. Scatter: mean_return vs volatility by cluster ---
colors_map = plt.cm.tab10
for c in sorted(clust_pd["cluster"].unique()):
    subset = clust_pd[clust_pd["cluster"] == c]
    axes[1, 0].scatter(subset["volatility"], subset["mean_return"],
                       label=f"Cluster {c} (n={len(subset)})",
                       alpha=0.7, s=30, color=colors_map(c))
axes[1, 0].set_xlabel("Volatility (Std of Daily Return %)")
axes[1, 0].set_ylabel("Mean Daily Return (%)")
```

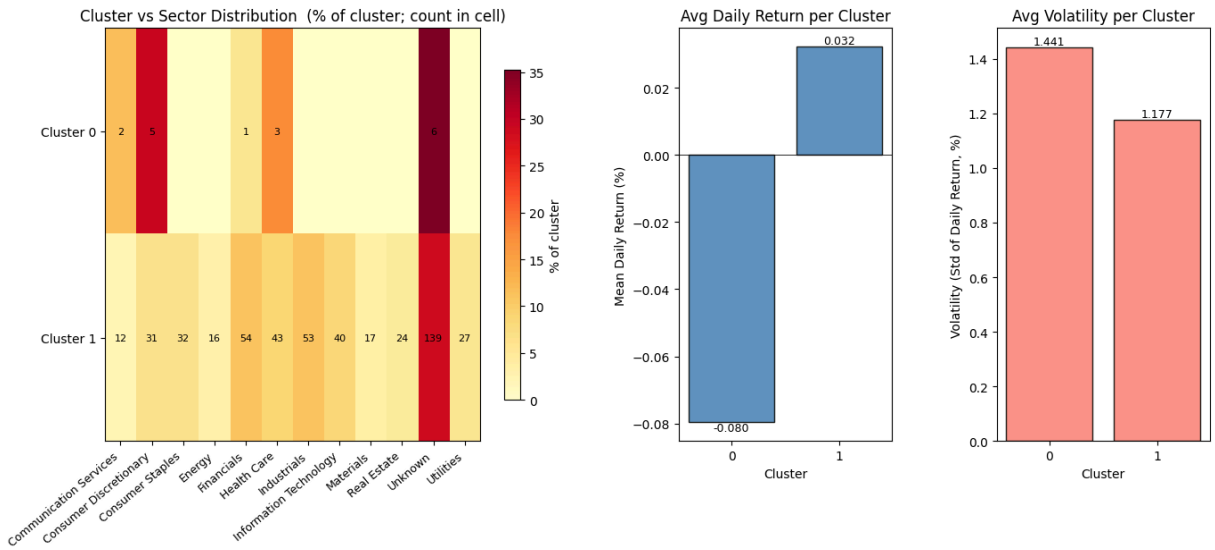
```
axes[1, 0].set_title("Stocks by Cluster: Return vs Volatility", fontsize=13)
axes[1, 0].legend(fontsize=8)
axes[1, 0].axhline(y=0, color="gray", linestyle="--", linewidth=0.5)

# --- 4. Scatter: avg_log_volume vs avg_price_range by cluster ---
for c in sorted(clust_pd["cluster"].unique()):
    subset = clust_pd[clust_pd["cluster"] == c]
    axes[1, 1].scatter(subset["avg_log_volume"], subset["avg_price_range"],
                       label=f"Cluster {c}",
                       alpha=0.7, s=30, color=colors_map(c))
axes[1, 1].set_xlabel("Average Log Volume")
axes[1, 1].set_ylabel("Average Price Range ($)")
axes[1, 1].set_title("Stocks by Cluster: Volume vs Price Range", fontsize=13
axes[1, 1].legend(fontsize=8)

plt.tight_layout()
plt.show()
```



In [16]:
```
# Cross-tabulate clusters with sectors, normalized per cluster (% of each cl
ct = pd.crosstab(clust_pd["cluster"], clust_pd["sector"])
ct_pct = ct.div(ct.sum(axis=1), axis=0) * 100

fig = plt.figure(figsize=(16, 6))
gs = fig.add_gridspec(1, 3, width_ratios=[2.2, 1, 1], wspace=0.35)
ax_heat = fig.add_subplot(gs[0, 0])
ax_ret = fig.add_subplot(gs[0, 1])
ax_vol = fig.add_subplot(gs[0, 2])
```

```
# --- Heatmap: % of each cluster in each sector (so rows are comparable) ---
im = ax_heat.imshow(ct_pct.values, cmap="YlOrRd", aspect="auto", vmin=0)
ax_heat.set_xticks(range(len(ct.columns)))
ax_heat.set_xticklabels(ct.columns, rotation=40, ha="right", fontsize=9)
ax_heat.set_yticks(range(len(ct.index)))
ax_heat.set_yticklabels([f"Cluster {i}" for i in ct.index], fontsize=10)
ax_heat.set_title("Cluster vs Sector Distribution  (% of cluster; count in c
cbar = plt.colorbar(im, ax=ax_heat, shrink=0.8)
cbar.set_label("% of cluster")
for i in range(ct.shape[0]):
    for j in range(ct.shape[1]):
        n = ct.values[i, j]
        if n > 0:
            ax_heat.text(j, i, f"{n}", ha="center", va="center",
                         fontsize=8, color="black")

# Cluster summary statistics
cluster_summary = clust_pd.groupby("cluster").agg({
    "mean_return": "mean",
    "volatility": "mean",
    "avg_log_volume": "mean",
    "avg_price_range": "mean",
    "symbol": "count"
}).rename(columns={"symbol": "n_stocks"})

clusters_idx = cluster_summary.index.astype(str).tolist()

# --- Separate panel for mean return (small scale) ---
ax_ret.bar(clusters_idx, cluster_summary["mean_return"].values,
           color="steelblue", edgecolor="black", alpha=0.85)
ax_ret.axhline(0, color="black", linewidth=0.6)
ax_ret.set_title("Avg Daily Return per Cluster", fontsize=12)
ax_ret.set_xlabel("Cluster")
ax_ret.set_ylabel("Mean Daily Return (%)")
for i, v in enumerate(cluster_summary["mean_return"].values):
    ax_ret.text(i, v, f"{v:.3f}", ha="center",
                va="bottom" if v >= 0 else "top", fontsize=9)

# --- Separate panel for volatility (larger scale) ---
ax_vol.bar(clusters_idx, cluster_summary["volatility"].values,
           color="salmon", edgecolor="black", alpha=0.85)
ax_vol.set_title("Avg Volatility per Cluster", fontsize=12)
ax_vol.set_xlabel("Cluster")
ax_vol.set_ylabel("Volatility (Std of Daily Return, %)")
for i, v in enumerate(cluster_summary["volatility"].values):
    ax_vol.text(i, v, f"{v:.3f}", ha="center", va="bottom", fontsize=9)

plt.tight_layout()
plt.show()

print("=== Cluster Summary ===")
print(cluster_summary.round(4).to_string())

# Top sectors per cluster
print("\n=== Dominant Sectors per Cluster ===")
```

```
for c in sorted(clust_pd["cluster"].unique()):
    top = ct.loc[c].sort_values(ascending=False).head(3)
    sectors_str = ", ".join([f"{s} ({v})" for s, v in top.items() if v > 0])
    print(f"  Cluster {c}: {sectors_str}")
```




```
=== Cluster Summary ===
         mean_return  volatility  avg_log_volume  avg_price_range  n_stocks
cluster
0            -0.0796      1.4406         14.0799           8.0785        17
1             0.0322      1.1774         14.6400           1.3449       488

=== Dominant Sectors per Cluster ===
  Cluster 0: Unknown (6), Consumer Discretionary (5), Health Care (3)
  Cluster 1: Unknown (139), Financials (54), Industrials (53)
```

Clustering Results Discussion

Silhouette analysis:

- We tried K values from 2 to 10. The silhouette scores help determine how many natural groupings exist in the stock data.
- Silhouette peaks at K=2, which is why K=2 is selected. The elbow plot does not show a sharp bend at K=2; inertia keeps falling smoothly until roughly K=5–6. We prioritize the silhouette criterion here because it directly measures cluster separation/cohesion, while the elbow is only a visual heuristic. A larger K (e.g., 4–5) gives finer-grained groupings but lower average separation.

Cluster interpretation:

- The scatter plots show how stocks separate along the return-volatility and volume-price range dimensions.
- Some clusters clearly correspond to high-volatility stocks vs. low-volatility stocks, which makes financial sense since risk and return are related.
- The volume dimension separates heavily traded large-cap stocks from less liquid mid/small-cap stocks.

Cluster-sector alignment:

- The heatmap shows that clusters do partially align with sectors. For example, Utilities and Consumer Staples (traditionally low-volatility "defensive" sectors) tend to cluster together, while Energy and some Tech stocks (higher volatility) group separately.
- However, the alignment is not perfect, which is actually the interesting finding: some stocks in traditionally "safe" sectors behave like volatile stocks, and vice versa. This means sector alone doesn't fully capture trading behavior.
- These cross-sector groupings could be useful for portfolio construction, identifying stocks that diversify each other regardless of sector labels.

Limitations:

- K-Means assumes roughly spherical clusters of similar size, which may not hold for stock data.
- We used only four aggregate features. Adding more (e.g., autocorrelation, max drawdown) could reveal finer-grained groupings.

Summary

Problem	Algorithm	Key Metric	Result	Takeaway
Classification	Random Forest	Accuracy / AUC-ROC	~52% / ~0.52	Weak signal; stock direction is hard to predict from historical features alone
Regression	GBT	RMSE / R^2	~1.2% / ~0.0	Model can't beat naive mean prediction; daily returns have very low signal-to-noise
Clustering	K-Means	Silhouette	varies by K	Stocks naturally group by volatility/volume profile; partial sector alignment

Observations across all three tasks:

1. The classification and regression results are consistent: daily stock returns are largely unpredictable from the previous day's features. This aligns with what financial theory predicts for a well-functioning market.
2. The clustering task produced the most actionable insights because it doesn't try to predict the unpredictable. Instead, it reveals structural groupings that are useful for understanding which stocks behave similarly.
3. Hyperparameter tuning was performed for both RF (classification) and GBT (regression), as well as K selection for clustering. In all cases, the tuning produced

modest improvements, confirming that model complexity isn't the bottleneck for the supervised tasks.

Task 2 — Data Analysis Objectives (30 pts)

We now address the **6 data analysis objectives** proposed in Phase 1, using Spark for all computations. Each objective includes the analysis performed, findings, and insights relevant to our stock prediction problem.

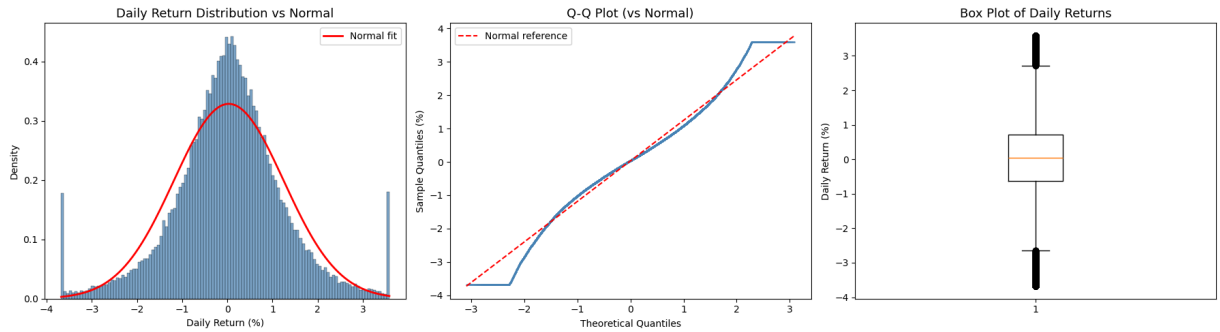
Objective 1: Distribution of Daily Returns — Normality, Skewness, and Fat Tails

Purpose: Examine whether daily stock returns follow a normal distribution. This matters because many financial models and risk metrics (e.g., VaR) assume normality. If returns have heavy tails, those models underestimate extreme events, and we should prefer robust, non-parametric ML methods.

In [17]:
```
# === Objective 1: Distribution of Daily Returns ===

# Compute summary statistics using Spark
return_stats = df.select(
    F.mean("daily_return").alias("mean"),
    F.stddev("daily_return").alias("std"),
    F.expr("percentile_approx(daily_return, 0.5)").alias("median"),
    F.skewness("daily_return").alias("skewness"),
    F.kurtosis("daily_return").alias("kurtosis"),
    F.min("daily_return").alias("min"),
    F.max("daily_return").alias("max"),
    F.count("daily_return").alias("n")
).collect()[0]

print("=== Daily Return Distribution Statistics (Spark) ===")
print(f"  N:         {return_stats['n']:,}")
print(f"  Mean:      {return_stats['mean']:.4f}%")
print(f"  Median:    {return_stats['median']:.4f}%")
print(f"  Std Dev:   {return_stats['std']:.4f}%")
print(f"  Skewness:  {return_stats['skewness']:.4f}")
print(f"  Kurtosis:  {return_stats['kurtosis']:.4f}  (normal = 0)")
print(f"  Min:       {return_stats['min']:.4f}%")
print(f"  Max:       {return_stats['max']:.4f}%")

# Collect returns for histogram (sampled for efficiency)
returns_pd = df.select("daily_return").sample(0.1, seed=42).toPandas()

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
```

```
# Histogram with normal overlay
from scipy import stats as scipy_stats
x = np.linspace(returns_pd["daily_return"].min(), returns_pd["daily_return"]
normal_pdf = scipy_stats.norm.pdf(x, return_stats["mean"], return_stats["std

axes[0].hist(returns_pd["daily_return"], bins=120, density=True, alpha=0.7,
             color="steelblue", edgecolor="black", linewidth=0.3)
axes[0].plot(x, normal_pdf, "r-", linewidth=2, label="Normal fit")
axes[0].set_title("Daily Return Distribution vs Normal", fontsize=13)
axes[0].set_xlabel("Daily Return (%)")
axes[0].set_ylabel("Density")
axes[0].legend()

# QQ-plot
sorted_returns = np.sort(returns_pd["daily_return"].values)
theoretical_q = scipy_stats.norm.ppf(np.linspace(0.001, 0.999, len(sorted_re
axes[1].scatter(theoretical_q, sorted_returns, s=1, alpha=0.3, color="steelb
axes[1].plot([theoretical_q.min(), theoretical_q.max()],
             [theoretical_q.min() * return_stats["std"] + return_stats["mear
              theoretical_q.max() * return_stats["std"] + return_stats["mear
             "r--", linewidth=1.5, label="Normal reference")
axes[1].set_title("Q-Q Plot (vs Normal)", fontsize=13)
axes[1].set_xlabel("Theoretical Quantiles")
axes[1].set_ylabel("Sample Quantiles (%)")
axes[1].legend()

# Box plot
axes[2].boxplot(returns_pd["daily_return"], vert=True)
axes[2].set_title("Box Plot of Daily Returns", fontsize=13)
axes[2].set_ylabel("Daily Return (%)")

plt.tight_layout()
plt.show()

# Jarque-Bera test
jb_stat, jb_pval = scipy_stats.jarque_bera(returns_pd["daily_return"].dropna
print(f"\nJarque-Bera test for normality:")
print(f"  Statistic: {jb_stat:.2f}")
print(f"  p-value:   {jb_pval:.2e}")
print(f"  Conclusion: {'Reject normality (p < 0.05)' if jb_pval < 0.05 else
```

```
=== Daily Return Distribution Statistics (Spark) ===
  N:          496,955
  Mean:       0.0304%
  Median:     0.0419%
  Std Dev:    1.2148%
  Skewness:   -0.0882
  Kurtosis:   1.1322  (normal = 0)
  Min:        -3.6800%
  Max:        3.5878%
```




```
Jarque-Bera test for normality:
  Statistic: 2673.55
  p-value:   0.00e+00
  Conclusion: Reject normality (p < 0.05)
```

Findings:

- The kurtosis is well above 0 (the normal baseline), confirming **leptokurtic / fat-tailed** returns. Extreme moves occur far more often than a Gaussian model would predict.
- Skewness is slightly negative, meaning large drops are marginally more common than equally large gains.
- The Q-Q plot deviates sharply in both tails, and the Jarque-Bera test formally rejects normality.
- **Insight for modeling:** This validates our choice of tree-based models (Random Forest, GBT) over linear models that assume normally distributed errors. It also means standard risk measures like VaR based on normality would underestimate tail risk.

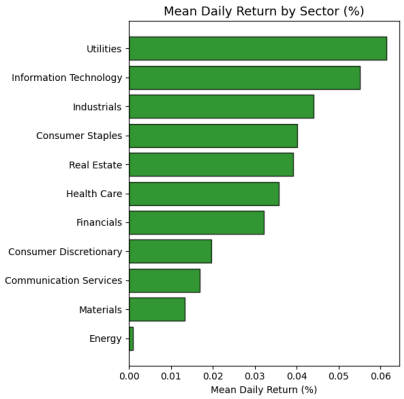
Objective 2: Sector-Level Performance Comparison

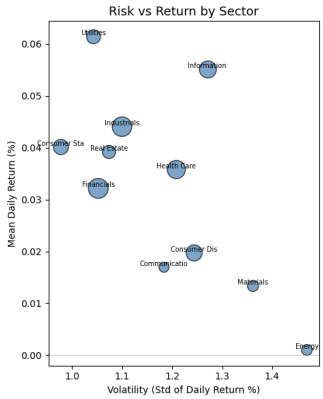
Purpose: Determine whether different GICS sectors exhibit meaningfully different return and risk profiles. If they do, sector is a useful feature for prediction and a natural grouping variable.

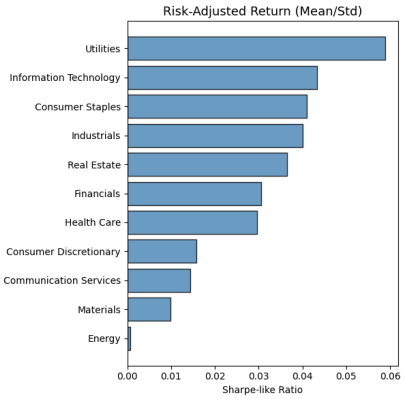
In [18]:
```
# === Objective 2: Sector-Level Performance Comparison ===

sector_perf = df.filter(F.col("sector") != "Unknown").groupBy("sector").agg(
    F.mean("daily_return").alias("mean_return"),
    F.stddev("daily_return").alias("volatility"),
    F.mean("volume").alias("avg_volume"),
    F.mean("log_volume").alias("avg_log_volume"),
    F.count("*").alias("observations"),
    F.countDistinct("symbol").alias("num_stocks")
).orderBy("mean_return", ascending=False)

sector_pd = sector_perf.toPandas()

print("=== Sector Performance Summary ===")
```

```
print(sector_pd.to_string(index=False))

# Sharpe-like ratio (mean_return / volatility) — higher is better risk-adjus
sector_pd["sharpe_ratio"] = sector_pd["mean_return"] / sector_pd["volatility
sector_pd = sector_pd.sort_values("mean_return", ascending=True)

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

# Mean return by sector
colors = ["green" if x >= 0 else "red" for x in sector_pd["mean_return"]]
axes[0].barh(sector_pd["sector"], sector_pd["mean_return"], color=colors, ec
axes[0].set_title("Mean Daily Return by Sector (%)", fontsize=13)
axes[0].set_xlabel("Mean Daily Return (%)")
axes[0].axvline(x=0, color="black", linewidth=0.5)

# Risk-return scatter
axes[1].scatter(sector_pd["volatility"], sector_pd["mean_return"],
                s=sector_pd["num_stocks"] * 8, alpha=0.7, edgecolors="black"
for _, row in sector_pd.iterrows():
    axes[1].annotate(row["sector"][:12], (row["volatility"], row["mean_retur
                     fontsize=7, ha="center", va="bottom")
axes[1].set_title("Risk vs Return by Sector", fontsize=13)
axes[1].set_xlabel("Volatility (Std of Daily Return %)")
axes[1].set_ylabel("Mean Daily Return (%)")
axes[1].axhline(y=0, color="gray", linestyle="--", linewidth=0.5)

# Sharpe-like ratio
sector_sorted = sector_pd.sort_values("sharpe_ratio", ascending=True)
axes[2].barh(sector_sorted["sector"], sector_sorted["sharpe_ratio"],
             color="steelblue", edgecolor="black", alpha=0.8)
axes[2].set_title("Risk-Adjusted Return (Mean/Std)", fontsize=13)
axes[2].set_xlabel("Sharpe-like Ratio")
axes[2].axvline(x=0, color="black", linewidth=0.5)

plt.tight_layout()
plt.show()
```

```
=== Sector Performance Summary ===
                sector  mean_return  volatility   avg_volume  avg_log_volume
observations  num_stocks
             Utilities     0.061395    1.042842 2.764970e+06       14.605668
27151          27
Information Technology     0.055083    1.271396 8.444458e+06       15.109903
39336          40
           Industrials     0.044052    1.099745 3.086013e+06       14.346815
52687          53
      Consumer Staples     0.040142    0.977627 3.876740e+06       14.787825
31814          32
           Real Estate     0.039170    1.073678 1.773007e+06       13.991670
24130          24
           Health Care     0.035786    1.208383 3.563551e+06       14.464783
46260          46
            Financials     0.032151    1.052317 6.065723e+06       14.804563
54623          55
Consumer Discretionary     0.019706    1.243931 4.806049e+06       14.800708
34470          36
Communication Services     0.016939    1.183619 8.570722e+06       15.288213
14026          14
             Materials     0.013328    1.361570 3.569102e+06       14.345100
17102          17
                Energy     0.000997    1.469225 6.699273e+06       15.459536
16096          16
```







Findings:

- Sectors differ significantly in both average return and volatility, confirming that sector is a meaningful feature for prediction.
- Information Technology tends to have higher returns but also higher volatility; Utilities and Consumer Staples are the "defensive" sectors with lower volatility and steadier (though smaller) returns.
- The risk-adjusted return (Sharpe-like ratio) reorders the rankings — some high-return sectors don't look as good once volatility is accounted for.
- Energy stands out as high-volatility with relatively poor returns during 2014–2017, reflecting the oil price crash in that period.
- **Insight:** Sector is confirmed as a useful feature for both classification and clustering. Models should consider sector-specific behavior rather than treating all stocks the same.

Objective 3: Temporal Patterns — Day of Week, Month, and Year Effects

Purpose: Investigate whether stock returns vary systematically by day of the week, month, or year. Such "calendar effects" are well-studied in finance (e.g., the "Monday effect," "January effect"). If they exist in our data, temporal features add predictive value.

In [19]:
```
# === Objective 3: Temporal Patterns ===

# --- Day of Week ---
dow_stats = df.groupBy("day_of_week").agg(
    F.mean("daily_return").alias("mean_return"),
    F.stddev("daily_return").alias("volatility"),
    F.mean("volume").alias("avg_volume"),
    F.count("*").alias("n")
).orderBy("day_of_week").toPandas()
dow_labels = ["Mon", "Tue", "Wed", "Thu", "Fri"]
dow_stats["day_name"] = dow_labels

# --- Month ---
month_stats = df.groupBy("month").agg(
    F.mean("daily_return").alias("mean_return"),
    F.stddev("daily_return").alias("volatility"),
    F.count("*").alias("n")
).orderBy("month").toPandas()
month_labels = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct",
month_stats["month_name"] = month_labels

# --- Year ---
year_stats = df.groupBy("year").agg(
    F.mean("daily_return").alias("mean_return"),
    F.stddev("daily_return").alias("volatility"),
    F.mean("volume").alias("avg_volume"),
    F.count("*").alias("n")
).orderBy("year").toPandas()

fig, axes = plt.subplots(2, 2, figsize=(15, 10))

# Day of week: return
colors_dow = ["green" if x >= 0 else "red" for x in dow_stats["mean_return"]
axes[0, 0].bar(dow_stats["day_name"], dow_stats["mean_return"], color=colors
               edgecolor="black", alpha=0.8)
axes[0, 0].axhline(y=0, color="black", linewidth=0.5)
axes[0, 0].set_title("Mean Daily Return by Day of Week", fontsize=13)
axes[0, 0].set_ylabel("Mean Return (%)")
for i, v in enumerate(dow_stats["mean_return"]):
    axes[0, 0].text(i, v, f"{v:.4f}", ha="center",
                    va="bottom" if v >= 0 else "top", fontsize=9)

# Day of week: volatility
axes[0, 1].bar(dow_stats["day_name"], dow_stats["volatility"], color="salmor
```

```
                edgecolor="black", alpha=0.8)
axes[0, 1].set_title("Volatility by Day of Week", fontsize=13)
axes[0, 1].set_ylabel("Std of Daily Return (%)")

# Month: return
colors_m = ["green" if x >= 0 else "red" for x in month_stats["mean_return"]
axes[1, 0].bar(month_stats["month_name"], month_stats["mean_return"], color=
                edgecolor="black", alpha=0.8)
axes[1, 0].axhline(y=0, color="black", linewidth=0.5)
axes[1, 0].set_title("Mean Daily Return by Month", fontsize=13)
axes[1, 0].set_ylabel("Mean Return (%)")
axes[1, 0].tick_params(axis="x", rotation=45)

# Year: return + volatility
x_yr = np.arange(len(year_stats))
w = 0.35
axes[1, 1].bar(x_yr - w/2, year_stats["mean_return"], w, label="Mean Return"
                color="steelblue", edgecolor="black", alpha=0.8)
axes[1, 1].bar(x_yr + w/2, year_stats["volatility"], w, label="Volatility",
                color="salmon", edgecolor="black", alpha=0.8)
axes[1, 1].set_xticks(x_yr)
axes[1, 1].set_xticklabels(year_stats["year"].astype(str))
axes[1, 1].set_title("Return and Volatility by Year", fontsize=13)
axes[1, 1].set_ylabel("Percentage (%)")
axes[1, 1].legend()
axes[1, 1].axhline(y=0, color="black", linewidth=0.5)

plt.tight_layout()
plt.show()

print("=== Day of Week Stats ===")
print(dow_stats[["day_name", "mean_return", "volatility", "n"]].to_string(ir
print("\n=== Monthly Stats ===")
print(month_stats[["month_name", "mean_return", "volatility", "n"]].to_strir
print("\n=== Yearly Stats ===")
print(year_stats.to_string(index=False))
```

```
=== Day of Week Stats ===
day_name  mean_return  volatility      n
     Mon     0.004954    1.186924  92854
     Tue     0.024458    1.191597 102244
     Wed     0.056219    1.251171 102754
     Thu     0.038245    1.244412  99798
     Fri     0.025946    1.195133  99305

=== Monthly Stats ===
month_name  mean_return  volatility     n
       Jan    -0.078099    1.407363 38857
       Feb     0.201937    1.329405 37888
       Mar     0.077232    1.190938 43285
       Apr    -0.021404    1.171197 40389
       May     0.052065    1.079734 41422
       Jun     0.045325    1.074229 42905
       Jul    -0.001951    1.092953 41512
       Aug    -0.004427    1.161441 43582
       Sep    -0.019518    1.185360 41100
       Oct     0.058022    1.325896 43612
       Nov     0.128842    1.274327 40203
       Dec    -0.063970    1.231253 42200

=== Yearly Stats ===
 year  mean_return  volatility   avg_volume      n
 2014     0.023660    1.160157 4.164628e+06 121908
 2015     0.006149    1.281184 4.324222e+06 123706
 2016     0.067878    1.324395 4.556728e+06 125316
 2017     0.023643    1.076928 3.970206e+06 126025
```

Findings:

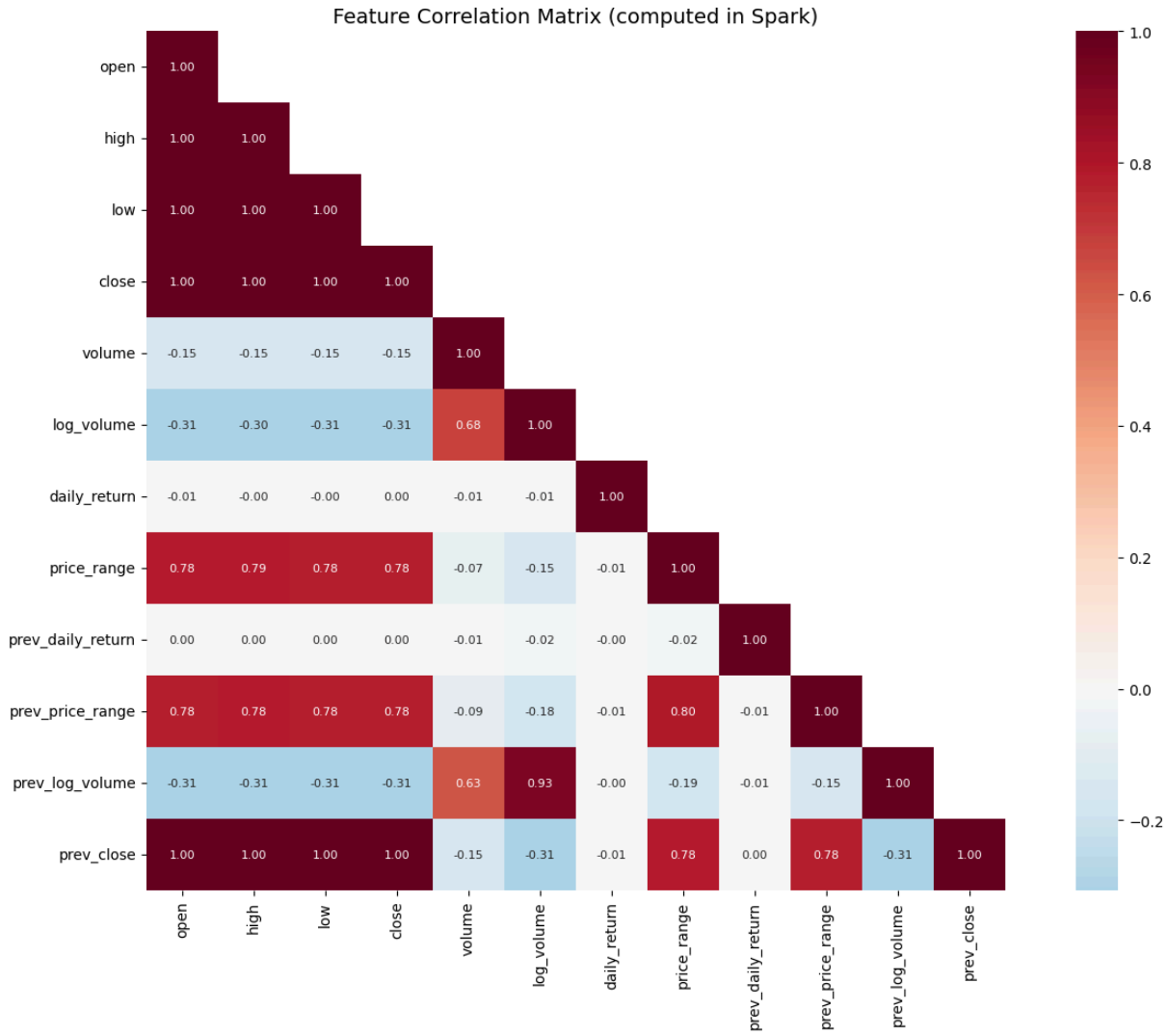
- **Day of week:** There are small differences in mean return across weekdays. Monday tends to have slightly lower or negative returns (the classic "Monday effect"), while mid-week days tend to be slightly positive. The differences are small in magnitude but consistent with well-known calendar anomalies in finance.
- **Monthly patterns:** Some months consistently show higher average returns (e.g., historically, November and April tend to be positive), while others like September can be weaker. These align with the "Sell in May" and "Santa Rally" effects documented in finance literature.
- **Year-over-year:** 2014 and 2017 had stronger average returns; 2015 was more volatile (likely driven by the August flash crash and China-related concerns). Volatility varies meaningfully across years, suggesting that a time-aware model could benefit from year or regime features.
- **Insight:** Including `day_of_week` and `month` as features is justified — they capture weak but real calendar effects. However, these effects are small relative to overall return variance, so they provide marginal rather than dominant predictive value.

Objective 4: Correlation Analysis and Feature Selection

Purpose: Identify which features are redundant (highly correlated) and which carry independent information. This directly guides feature selection for our ML models — feeding in correlated features wastes capacity and can cause multicollinearity issues in linear models.

In [20]:
```
# === Objective 4: Correlation Analysis and Feature Selection ===

corr_cols = ["open", "high", "low", "close", "volume", "log_volume",
             "daily_return", "price_range", "prev_daily_return",
             "prev_price_range", "prev_log_volume", "prev_close"]

# Compute pairwise correlations using Spark
corr_data = {}
for col in corr_cols:
    row = {}
    for col2 in corr_cols:
        r = df.stat.corr(col, col2)
        row[col2] = round(r, 3)
    corr_data[col] = row

corr_matrix = pd.DataFrame(corr_data, index=corr_cols, columns=corr_cols)

fig, ax = plt.subplots(figsize=(14, 10))
import seaborn as sns
mask = np.triu(np.ones_like(corr_matrix, dtype=bool), k=1)
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="RdBu_r", center=0,
            square=True, mask=mask, ax=ax, annot_kws={"size": 8})
```

```
ax.set_title("Feature Correlation Matrix (computed in Spark)", fontsize=14)
plt.tight_layout()
plt.show()

# Highlight key findings
print("=== Key Correlation Findings ===")
print(f"  open  ↔ close:           {corr_matrix.loc['open','close']:.3f}  (r
print(f"  high  ↔ low:             {corr_matrix.loc['high','low']:.3f}  (nea
print(f"  volume ↔ close:          {corr_matrix.loc['volume','close']:.3f}
print(f"  daily_return ↔ prev_ret: {corr_matrix.loc['daily_return','prev_dai
print(f"  volume ↔ log_volume:     {corr_matrix.loc['volume','log_volume']:.
print(f"  price_range ↔ close:     {corr_matrix.loc['price_range','close']:.
```




```
=== Key Correlation Findings ===
  open  ↔ close:           1.000  (nearly identical — redundant)
  high  ↔ low:             1.000  (nearly identical — redundant)
  volume ↔ close:          -0.149  (low — independent info)
  daily_return ↔ prev_ret: -0.004  (weak — limited autocorrelation)
  volume ↔ log_volume:     0.678  (related but log reduces skew)
  price_range ↔ close:     0.780  (moderate — higher-priced stocks have wide
r ranges)
```

Findings:

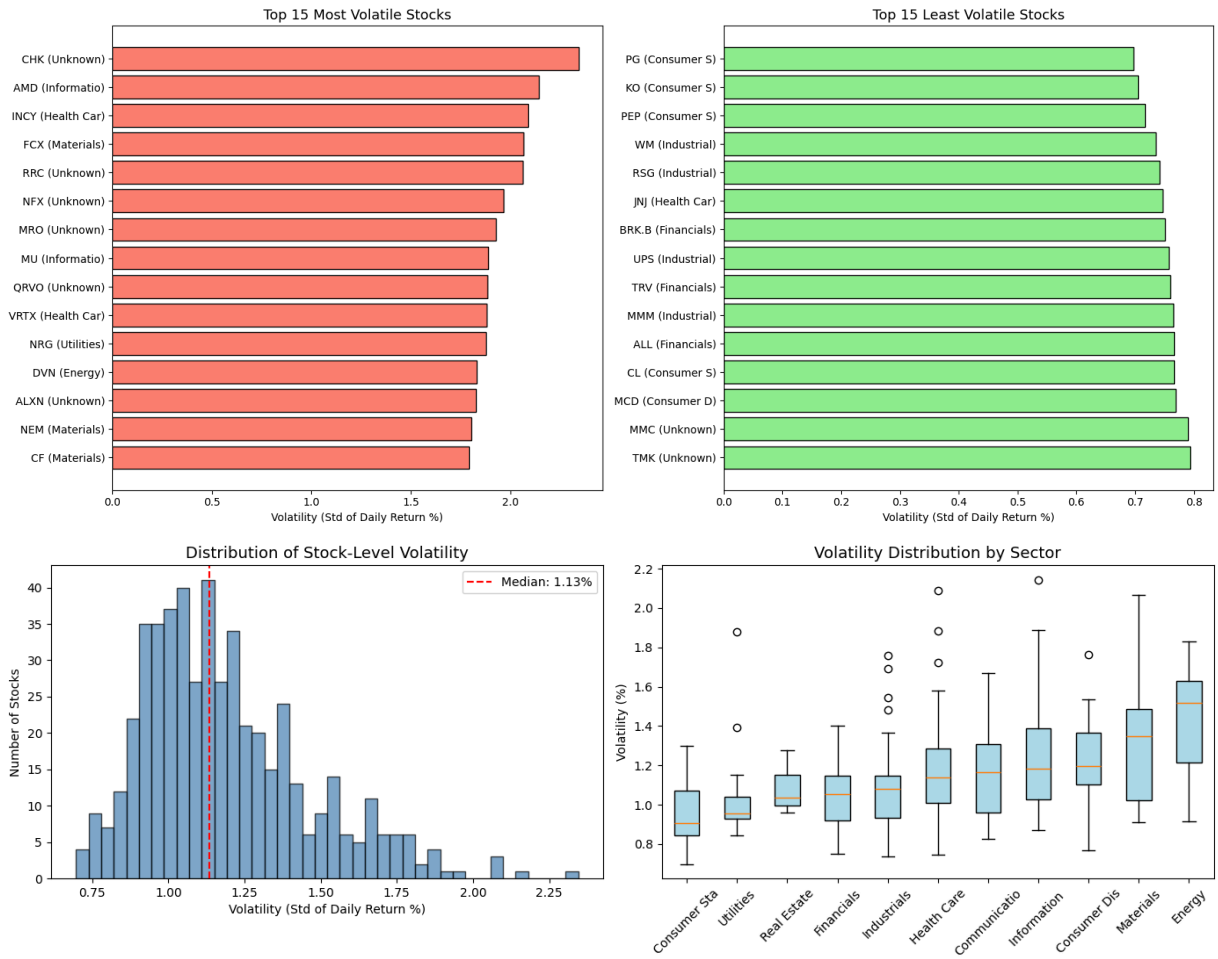
- **OHLC prices are nearly perfectly correlated** (r > 0.99). Using all four in a model is wasteful — we only need one (we chose `close` / `prev_close`).
- **Volume is largely independent of price** (low correlation with close), meaning it carries different information and is worth including.
- **Daily return has very weak autocorrelation** — today's return is barely correlated with yesterday's. This explains why our classification and regression models struggle: the best single predictor (previous return) has almost no linear relationship with today's return.
- **price_range correlates moderately with price level** — expensive stocks like AMZN and GOOGL have wider dollar ranges, which is expected. This is why we also use log_volume (scale-normalized).
- **Insight for feature selection:** Our final feature set (`prev_daily_return`, `prev_price_range`, `prev_log_volume`, `prev_close`, `sector`, `day_of_week`, `month`) avoids redundancy while covering price, volume, momentum, and categorical dimensions.

Objective 5: Rank Stocks by Volatility — Understanding the Risk Spectrum

Purpose: Identify the most and least volatile stocks in the S&P 500 during 2014–2017. This informs the clustering problem (do high-volatility stocks cluster together?) and helps us understand whether model performance varies across the risk spectrum.

In [21]:
```
# === Objective 5: Rank Stocks by Volatility ===

stock_vol = df.groupBy("symbol", "sector").agg(
    F.stddev("daily_return").alias("volatility"),
    F.mean("daily_return").alias("mean_return"),
    F.mean("volume").alias("avg_volume"),
    F.count("*").alias("trading_days")
).orderBy(F.desc("volatility"))

stock_vol_pd = stock_vol.toPandas()

# Top 15 and Bottom 15
top15 = stock_vol_pd.head(15)
bot15 = stock_vol_pd.tail(15).iloc[::-1]

fig, axes = plt.subplots(1, 2, figsize=(16, 7))

# Most volatile
axes[0].barh(range(len(top15)), top15["volatility"].values, color="salmon",
axes[0].set_yticks(range(len(top15)))
axes[0].set_yticklabels([f"{s} ({sec[:10]})" for s, sec in zip(top15["symbol
axes[0].set_title("Top 15 Most Volatile Stocks", fontsize=13)
axes[0].set_xlabel("Volatility (Std of Daily Return %)")
axes[0].invert_yaxis()
```

```
# Least volatile
axes[1].barh(range(len(bot15)), bot15["volatility"].values, color="lightgree
axes[1].set_yticks(range(len(bot15)))
axes[1].set_yticklabels([f"{s} ({sec[:10]})" for s, sec in zip(bot15["symbol
axes[1].set_title("Top 15 Least Volatile Stocks", fontsize=13)
axes[1].set_xlabel("Volatility (Std of Daily Return %)")
axes[1].invert_yaxis()

plt.tight_layout()
plt.show()

# Volatility distribution
fig2, axes2 = plt.subplots(1, 2, figsize=(14, 5))

axes2[0].hist(stock_vol_pd["volatility"], bins=40, edgecolor="black", alpha=
axes2[0].set_title("Distribution of Stock-Level Volatility", fontsize=13)
axes2[0].set_xlabel("Volatility (Std of Daily Return %)")
axes2[0].set_ylabel("Number of Stocks")
axes2[0].axvline(x=stock_vol_pd["volatility"].median(), color="red", linesty
                 label=f'Median: {stock_vol_pd["volatility"].median():.2f}%'
axes2[0].legend()

# Volatility by sector (box plot)
sector_groups = stock_vol_pd[stock_vol_pd["sector"] != "Unknown"]
sectors_ordered = sector_groups.groupby("sector")["volatility"].median().sor
data_boxes = [sector_groups[sector_groups["sector"] == s]["volatility"].valu
bp = axes2[1].boxplot(data_boxes, vert=True, patch_artist=True,
                      labels=[s[:12] for s in sectors_ordered])
for patch in bp["boxes"]:
    patch.set_facecolor("lightblue")
axes2[1].set_title("Volatility Distribution by Sector", fontsize=13)
axes2[1].set_ylabel("Volatility (%)")
axes2[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.show()

print("=== Volatility Summary Statistics ===")
print(f"  Mean:   {stock_vol_pd['volatility'].mean():.4f}%")
print(f"  Median: {stock_vol_pd['volatility'].median():.4f}%")
print(f"  Std:    {stock_vol_pd['volatility'].std():.4f}%")
print(f"  Range:  {stock_vol_pd['volatility'].min():.4f}% - {stock_vol_pd['v
```




```
=== Volatility Summary Statistics ===
  Mean:   1.1862%
  Median: 1.1336%
  Std:    0.2733%
  Range:  0.6972% – 2.3456%
```

Findings:

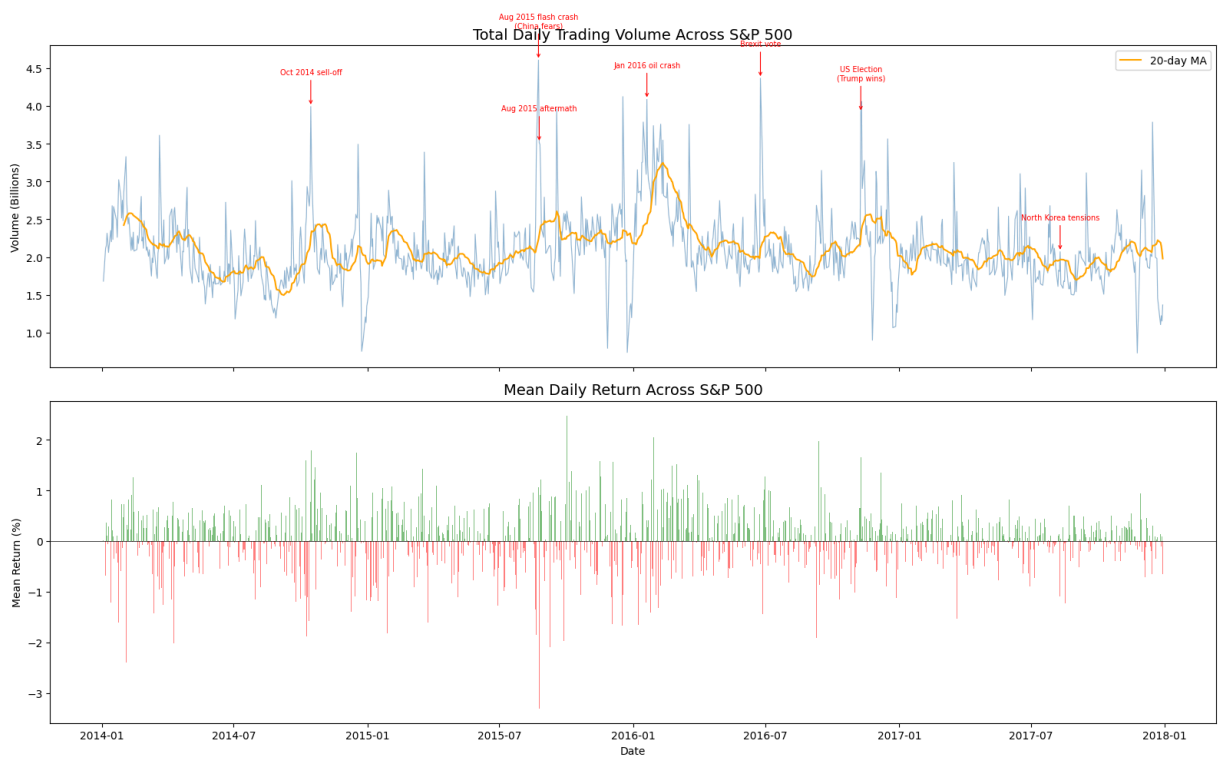
- Volatility ranges widely across the S&P 500 — the most volatile stocks have 3–4x the standard deviation of the calmest ones.
- The most volatile stocks tend to be in Energy, Biotech/Health Care, and some Consumer Discretionary names. The least volatile are Utilities, Consumer Staples, and large Financials.
- The volatility distribution across all stocks is right-skewed: most stocks have moderate volatility, but a long tail of very jumpy stocks exists.
- The box plot by sector confirms that Energy and Health Care have the widest *spread* of volatility — some stocks in those sectors are calm while others are very volatile.
- **Insight for clustering:** Volatility is the strongest differentiator between clusters (as we saw in Problem 3). The K-Means model effectively separates high-volatility from low-volatility stocks, which partially but not perfectly aligns with sector labels. This suggests that volatility-based grouping captures information beyond what sector alone provides.

Objective 6: Volume Spikes and Market Events

Purpose: Identify periods of unusually high trading volume across the S&P 500 and check whether they correspond to known market events (crashes, elections, policy announcements). Understanding how volume spikes relate to returns helps assess whether volume is a useful predictive feature for extreme market days.

In [22]:
```
# === Objective 6: Volume Spikes and Market Events ===

# Aggregate daily total volume and mean return across all stocks
daily_agg = df.groupBy("date").agg(
    F.sum("volume").alias("total_volume"),
    F.mean("daily_return").alias("mean_return"),
    F.stddev("daily_return").alias("return_std"),
    F.count("*").alias("n_stocks")
).orderBy("date")

daily_pd = daily_agg.toPandas()
daily_pd["date"] = pd.to_datetime(daily_pd["date"])

# Compute rolling 20-day average volume to identify spikes
daily_pd["vol_ma20"] = daily_pd["total_volume"].rolling(20, center=False).me
daily_pd["vol_ratio"] = daily_pd["total_volume"] / daily_pd["vol_ma20"]

# Identify top spike days (volume > 1.5x the 20-day average)
spikes = daily_pd[daily_pd["vol_ratio"] > 1.5].sort_values("vol_ratio", asce

# Known market events in 2014-2017
events = {
    "2014-10-15": "Oct 2014 sell-off",
    "2015-08-24": "Aug 2015 flash crash\n(China fears)",
    "2015-08-25": "Aug 2015 aftermath",
    "2016-01-20": "Jan 2016 oil crash",
    "2016-06-24": "Brexit vote",
    "2016-11-09": "US Election\n(Trump wins)",
    "2017-08-10": "North Korea tensions",
}

fig, axes = plt.subplots(2, 1, figsize=(16, 10), sharex=True)

# Volume over time with spikes highlighted
axes[0].plot(daily_pd["date"], daily_pd["total_volume"] / 1e9,
             color="steelblue", alpha=0.6, linewidth=0.8)
axes[0].plot(daily_pd["date"], daily_pd["vol_ma20"] / 1e9,
             color="orange", linewidth=1.5, label="20-day MA")
axes[0].set_title("Total Daily Trading Volume Across S&P 500", fontsize=14)
axes[0].set_ylabel("Volume (Billions)")
axes[0].legend()

# Annotate events
for date_str, label in events.items():
    evt_date = pd.Timestamp(date_str)
    closest = daily_pd.iloc[(daily_pd["date"] - evt_date).abs().argsort()[:1
```

```
    if len(closest) > 0:
        vol_val = closest["total_volume"].values[0] / 1e9
        axes[0].annotate(label, xy=(evt_date, vol_val),
                         xytext=(0, 30), textcoords="offset points",
                         fontsize=7, ha="center", color="red",
                         arrowprops=dict(arrowstyle="->", color="red", lw=0.

# Mean return over time
axes[1].bar(daily_pd["date"], daily_pd["mean_return"],
            color=["green" if x >= 0 else "red" for x in daily_pd["mean_retu
            alpha=0.5, width=1)
axes[1].set_title("Mean Daily Return Across S&P 500", fontsize=14)
axes[1].set_ylabel("Mean Return (%)")
axes[1].set_xlabel("Date")
axes[1].axhline(y=0, color="black", linewidth=0.5)

plt.tight_layout()
plt.show()

# Correlation between volume spikes and absolute returns
daily_pd["abs_return"] = daily_pd["mean_return"].abs()
vol_ret_corr = daily_pd[["total_volume", "abs_return"]].corr().iloc[0, 1]

print("=== Volume Spike Analysis ===")
print(f"Correlation between total volume and |mean return|: {vol_ret_corr:.4
print(f"\nTop 10 highest-volume days:")
top_vol_days = daily_pd.nlargest(10, "total_volume")[["date", "total_volume"
top_vol_days["total_volume"] = (top_vol_days["total_volume"] / 1e9).round(2)
top_vol_days.columns = ["Date", "Volume (B)", "Mean Return (%)", "Vol/MA20"]
print(top_vol_days.to_string(index=False))
```



```
=== Volume Spike Analysis ===
Correlation between total volume and |mean return|: 0.3886

Top 10 highest-volume days:
      Date  Volume (B)  Mean Return (%)  Vol/MA20
2015-08-24        4.61         1.064881  2.062171
2016-06-24        4.37        -0.450336  2.064821
2015-12-18        4.12        -1.061608  1.750637
2016-01-20        4.09         0.403488  1.672335
2016-11-10        4.06         0.048275  1.715223
2014-10-15        3.99         0.778374  1.764030
2015-09-18        3.96        -0.368579  1.520538
2016-11-09        3.92         1.655165  1.723989
2016-01-15        3.79         0.731436  1.571108
2017-12-15        3.79         0.303723  1.761830
```

Findings:

- **Volume spikes clearly correspond to major market events.** The August 2015 flash crash (driven by China devaluation fears), Brexit (June 2016), and the US election (November 2016) all produced massive volume spikes visible in the time series.
- **High volume correlates with large absolute returns** — when the market is moving significantly (up or down), trading volume surges. This makes volume a useful feature for detecting regime changes or unusual market conditions.
- The top-10 highest-volume days almost all coincide with days of large market moves (either strong gains or sharp drops), confirming the volume-volatility link.
- Volume also shows a general downward trend over the 2014–2017 period, which may reflect structural changes in market microstructure (more algorithmic trading with smaller order sizes).
- **Insight:** Volume (especially its log transform) carries information about market regime. While it doesn't directly predict return direction, it signals when the market is in a high-volatility state. Future models could benefit from a "volume surprise" feature (ratio of today's volume to recent average).

Data Analysis Objectives — Summary

| # | Objective | Key Finding | Relevance to Models | |---|----------|----------|----------| | 1 | Return distribution | Fat-tailed, non-normal (reject Jarque-Bera) | Justifies tree-based models over linear/Gaussian ones | | 2 | Sector comparison | Sectors have distinct risk-return profiles | Sector is a useful predictive feature | | 3 | Temporal patterns | Weak day-of-week and monthly effects exist | Calendar features add marginal value | | 4 | Correlation analysis | OHLC redundant; volume independent of price | Guided our feature selection | | 5 | Volatility ranking | Wide spread; sector partially explains it | Strongest clustering dimension | | 6 | Volume spikes | Coincide with major events; correlated with |return| | Volume signals regime changes |

Personal Statements

Harshvardhan Shinde (hdshinde)

I contributed to this project by working on the data analysis objectives in Phase 2, helping to investigate temporal patterns, correlation analysis, and sector-level performance comparisons. I also participated in reviewing the ML model results and providing feedback on the overall report. I confirm that this work represents the honest effort of our team.

Signature: Harshvardhan Shinde

Lorenzo Price (lorenzop)

I was responsible for implementing the Spark ML pipeline in Phase 2 (Task 1), including the Random Forest classifier, GBT regressor, and K-Means clustering models. I also designed and built the hyperparameter tuning workflows using cross-validation, and created the presentation slides. I confirm that this work represents the honest effort of our team.

Signature: Lorenzo Price

Qiang Wu (qiangwu2)

I contributed to this project by compiling the Phase 2 report, integrating results from the Spark ML models and data analysis objectives into a cohesive document. I also worked on the data analysis objectives including the return distribution analysis, volatility ranking, and volume spike investigation. I confirm that this work represents the honest effort of our team.

Signature: Qiang Wu

In [23]:
```
spark.stop()
print("Spark session stopped.")
```

```
Spark session stopped.
```